



About Time: Advances, Challenges, and Outlooks of Action Understanding

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Abstract

We have witnessed impressive advances in video action understanding. Increased dataset sizes, variability, and computation availability have enabled leaps in performance and task diversification. Current systems can provide coarse- and fine-grained descriptions of video scenes, extract segments corresponding to queries, synthesize unobserved parts of videos, and predict context across multiple modalities. This survey comprehensively reviews advances in uni- and multi-modal action understanding across a range of tasks. We focus on prevalent challenges, overview widely adopted datasets, and survey seminal works with an emphasis on recent advances. We broadly distinguish between three temporal scopes: (1) recognition tasks of actions observed in full, (2) prediction tasks for ongoing partially observed actions, and (3) forecasting tasks for subsequent unobserved action(s). This division allows us to identify specific action modeling and video representation challenges. Finally, we outline future directions to address current shortcomings.

Keywords Action Understanding · Action Recognition · Action Prediction · Action Anticipation

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1 Introduction

For decades, analyzing human actions in videos has been of particular interest to the computer vision community. Videos are prominent in both our social and professional lives. Over time, the analysis of actions has shifted from the well-understood task of action recognition towards the fundamental and broader area of action understanding. Shown in Figure 1, action understanding now includes diverse tasks based on prediction and anticipation with multimodal inputs. The unique challenges and novel computation paradigms are the core focus of our survey.

In developmental psychology, action understanding has been explored across several psychological aspects (Thompson et al., 2019):

The ability to understand the action performed relates to differentiating between analogous actions (Gallese et al., 1996; Jeannerod, 1994) and conceptualizing *how* an action is performed (Spunt et al., 2011).

Determining the goal of the action has been studied in the context of immediate goals (Calvo-Merino et al., 2005; Kohler et al., 2002; Rizzolatti et al., 2001) in relation to motor functions for the execution of actions and the sensory perception of actions performed by others.

Determining the actor's intention refers to identifying high-level goals and motivations to perform actions (Kilner, 2011). Intentions have been defined as the sequential

grouping of individual actions (Fogassi et al., 2005) and their abstract associated target (Uithol et al., 2011).

1.1 Taxonomy of this survey

Inspired by the cognitive aspects of action understanding, we define three broad *temporal scopes* to group seminal machine vision action understanding tasks. We visualize action sequence progression in Figure 2 with a currently (partially) performed action followed by a subsequent action. Tasks that require an action to be *observed in full* are broadly referred to as **recognition** tasks and infer information such as the action categories or high-level semantics. **Predictions** about the ongoing actions are made from partial observations of actions *not yet completed*. **Forecasting** tasks use the currently observed action(s) to reason about future actions *not yet observed*. We discuss relevant previous surveys for each of these three temporal scopes and overview their focus in Table 1.

Recognition. As seen by the top rows in Table 1, early works on action recognition have primarily focused on motion modeling. Aggarwal et al. (1994) used a taxonomy of rigidity and subsequently (Aggarwal et al., 1998) introduced subdivisions based on prior knowledge of the object's shape. Cedras and Shah (1995) and later (Moeslund & Granum, 2001) discussed temporal modeling approaches in the context of classification and tracking. As overviewed by Buxton (2003), tracking has also been applied to more complex tasks such as behavior analysis or non-verbal human interactions. Subsequent overviews were more task-oriented, focusing on action classification and localization (Weinland et al., 2011), behavior understanding (Chaaaraoui et al., 2012), and surveillance applications (Vishwakarma & Agrawal, 2013) emerged based on later advancements. Simultaneously, Turaga et al. (2008) and Poppe (2010) discussed approaches addressing atomic actions and group activities. Herath et al. (2017) provided an initial summary of approaches using learned features for action recognition. Following surveys covered adaptations of deep learning approaches for topics such as depth-based motion recognition (Wang et al., 2018d), activity recognition (Beddiar et al., 2020), human-human interactions (Stergiou & Poppe, 2019), and pose estimation (Zheng et al., 2020a). Sun et al. (2022b) reviewed approaches across modalities, combining motion features, audio, and vision. More recently, Selva et al. (2023) discussed attention-based approaches for video tasks while Schiappa et al. (2023) focused on self-supervised (SSL) approaches. Madan et al. (2024) presented a comprehensive overview of language-enabled action understanding models, focusing on video foundation models.

Prediction. Recent advancements in action recognition have also sparked interest in predictive tasks from partial observations. Rasouli (2020) discussed four main domains of predic-

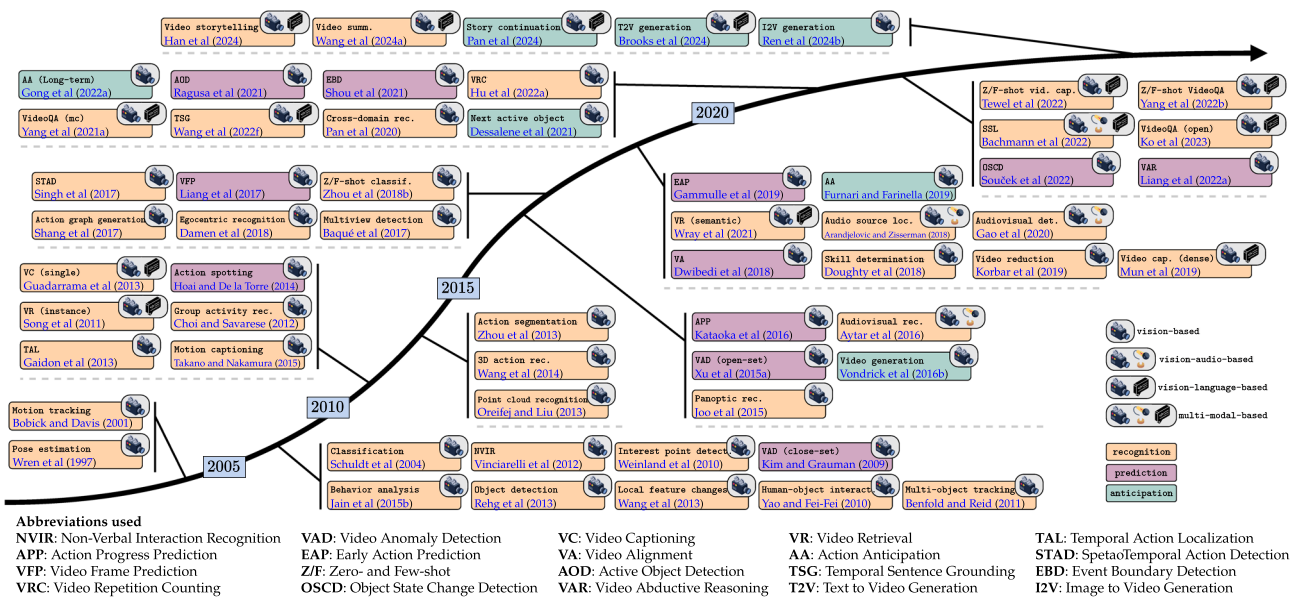


Fig. 1 Action understanding historical overview. We present popular tasks over time. Landmark papers are selected by their relevance to the period's trends. Most tasks remain popular today

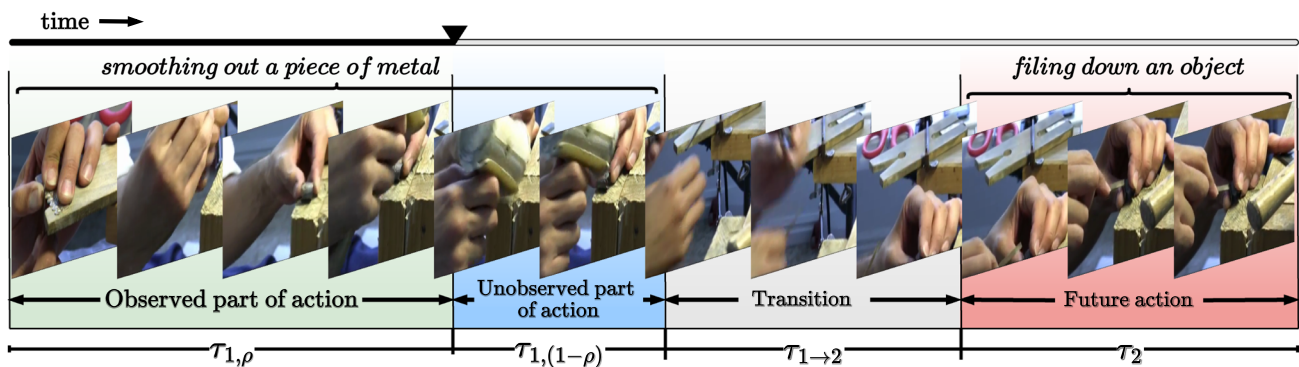


Fig. 2 Action understanding tasks. The progress of the video is indicated by the top bar. From the currently performed action of total duration τ_1 , only the $\tau_{1,\rho} < \tau_1$ part is readily observable. After a transition period $0 \leq \tau_{1 \rightarrow 2}$, another action is performed with duration τ_2 .

Action recognition tasks consider full observations of the action at τ_1 . Action prediction uses only part $\tau_{1,\rho}$ of the ongoing action. Action forecasting uses current action at τ_1 to predict future actions. Video example sourced from Wang et al. (2019b)

tive models including video, action, trajectory, and motion prediction. Kong and Fu (2022) described recent action recognition and prediction advancements. They focused on applications in domains such as robot vision, surveillance, and driver behavior prediction. The surveys of Dhiman and Vishwakarma (2019) and Ramachandra et al. (2020) overviewed predictive methods specifically for anomaly detection. As shown in Table 1 overviews on these tasks are scarce.

Forecasting. Action forecasting tasks have become prevalent parts of action understanding research. Rodin et al. (2021) discussed future action anticipation in egocentric videos. Zhong et al. (2023b) overviewed short and long-term action anticipation methods. Hu et al. (2022c) provided a

review of online and anticipation works. Recently, Plizzari et al. (2024) discussed challenges in egocentric videos and presented future directions for multiple tasks including forecasting.

Despite their extensive coverage, prior surveys focus on specific aspects of action understanding. As shown in Table 1, a critical overview that holistically explores action understanding is currently missing in the literature.

This survey fills this void by focusing on advancements across a broad range of action understanding tasks. We do this from a temporal perspective. We survey general approaches for modeling actions in videos over the years in Section 2, and discuss common datasets and benchmarks in Section 3. We then detail recognition tasks in Section 4, predictive tasks

Table 1 Action understanding surveys through the years. For each survey, we note the year and number of papers covered. We identify the coverage of temporal scopes, including recognition (Rec.), prediction (Pred.), and forecasting (For.). Broad objectives include multimodality (MM), self-supervision (SSL), and multi-view (MV). We further highlight other specific tasks discussed, such as human interactions (HI), long video understanding (LVU). Scopes/objectives/tasks addressed partially within surveys are denoted with (partial), and the main focus is denoted with ✓

Author(s)	Year	#Papers	Temporal Scope			Objectives			Tasks	
			Rec.	Pred.	For.	MM	SSL	MV	HI	LVU
Aggarwal et al. (1994)	1994	69	(partially)							
Cedras and Shah (1995)	1995	76	(partially)							
Aggarwal et al. (1998)	1998	104	(partially)							
Aggarwal and Cai (1999)	1999	51	(partially)							
Moeslund and Granum (2001)	2001	155	(partially)							
Buxton (2003)	2003	88	(partially)						✓	
Moeslund et al. (2006)	2006	424	✓			(partially)				
Yilmaz et al. (2006)	2006	160	(partially)						(partially)	
Turaga et al. (2008)	2008	144	✓		(partially)	(partially)			✓	
Poppe (2010)	2010	180	✓							
Weinland et al. (2011)	2011	153	✓			(partially)			✓	
Chaaroui et al. (2012)	2012	123	✓			(partially)		✓	✓	
Metaxas and Zhang (2013)	2013	188				(partially)		(partially)	✓	
Vishwakarma and Agrawal (2013)	2013	231	✓			(partially)				
Herath et al. (2017)	2017	161	✓							
Wang et al. (2018d)	2018	182	✓	(partially)		(partially)		(partially)	(partially)	
Dhiman and Vishwakarma (2019)	2019	208			✓	(partially)		(partially)	(partially)	
Hussain et al. (2019)	2019	141	✓			✓		(partially)	✓	
Stergiou and Poppe (2019)	2019	178	✓						✓	
Yao et al. (2019)	2019	106	✓							
Zhang et al. (2019b)	2019	127	✓							
Beddiar et al. (2020)	2020	237	✓		(partially)			(partially)	✓	
Ramachandra et al. (2020)	2020	109		✓						
Zheng et al. (2020a)	2020	317				✓				
Rasouli (2020)	2020	333		✓						
Pareek and Thakkar (2021)	2021	218	✓			(partially)				
Rodin et al. (2021)	2021	156			✓	(partially)		✓		✓
Song et al. (2021)	2021	157	✓							
Sun et al. (2022b)	2022	503	✓			(partially)	✓			
Kong and Fu (2022)	2022	337	✓	(partially)				✓	(partially)	

Table 1 continued

Author(s)	Year	#Papers	Temporal Scope		For.	Objectives		SSL	MV	Tasks	
			Rec.	Pred.		MM				HI	LVU
Hu et al. (2022c)	2022	168	✓		✓					✓	(partially)
Oprea et al. (2022)	2022	211	✓	✓							
Schiappa et al. (2023)	2023	216	✓					✓	✓		✓
Selva et al. (2023)	2023	209	✓								
Wang et al. (2023a)	2023	229	✓			(partially)			(partially)		
Zhong et al. (2023b)	2023	207	✓		✓	✓		✓	✓		✓
Ding et al. (2023)	2023	168	✓						✓		(partially)
Tang et al. (2023)	2023	338	✓			✓		✓	(partially)		✓
Plizzari et al. (2024)	2024	367	✓		✓	✓			✓	✓	✓
Madan et al. (2024)	2024	367	✓			✓		✓			✓
Lai et al. (2024c)	2024	202		(partially)	✓	✓			✓		✓
Stergiou and Poppe (this survey)	2025	1284	✓	✓	✓	✓		✓	✓	✓	✓

in Section 5, and forecasting tasks in Section 6. Based on the temporal scopes, we then outline the main challenges and provide future directions in Section 7. We conclude in Section 8.

2 Modeling actions in videos

In this section, we define two general groups of approaches for encoding videos without explicitly relating them to tasks. We start with characterizing key challenges in Section 2.1. Approaches discussed in Section 2.2 model spatial and temporal information separately, while works overviewed in Section 2.3 use joint spatiotemporal representations.

2.1 Challenges in action representation

The diversity of the video input poses several challenges. **Intra-class variations** in the visual appearance of actions of the same category across videos can be due to viewpoint, occlusions, background noise, or lighting conditions. The performances and durations of actions can also significantly deviate. Such variations appear across datasets (Grauman et al., 2022; Kay et al., 2017; Miech et al., 2019; Soomro et al., 2012). Training/test set instance distribution variance can also significantly impact the performance and overall generalization of the learned semantics. Challenging action instances can be traced to feature representations further from the training set distribution in such cases.

Since action understanding tasks are increasingly semantic, we also face challenges in the diversity and granularity of the target outputs. Interpretation of the visual input, and sometimes the lack of observable information, increasingly requires higher-level understanding. Consequently, the relation between visual input and model output becomes more complex. **Vocabulary limitations** present challenges as action categories are often finite. The generalization of models to *open-set* or *cross-domain* settings primarily depends on the similarity between seen and unseen instances. Limited *inter-class variation* further affects good representation performance of rare coarse-grained concepts of visually similar actions. This issue is more prevalent for tasks that require fine-grained semantic granularities.

2.2 Separating visual and temporal information

We first discuss approaches that process visual and temporal information independently.

Tracking and template matching. Early works (Bobick & Davis, 2001) have applied template matching to spatially and temporally localize motions. These approaches relied on view-specific representations of movements, in the form of templates, to capture underlying motion similar-

ity across action instances. Templates have been explored through local patches (Shechtman & Irani, 2005), correlation filters (Rodriguez et al., 2008), and voxels (Ke et al., 2007). Another line of research has considered temporal pattern discovery by directly tracking visual features over time (Cipolla & Blake, 1990; Isard & Blake, 1998; Rohr, 1994). Template approaches have relied on assumptions such as static backgrounds, fixed camera views, and linear motions that limit the exploration of intra-class variability.

Local descriptors. Motivated by the observation that actions can be characterized through appearance changes over time, a set of approaches aims to associate per-frame changes from local descriptor features to action categories. Pose primitives (Thurau & Hlaváč, 2008), temporal bins (Nowozin et al., 2007), pictorial structures (Tran et al., 2012), and graphical structures of the actions (Ni et al., 2014) have been explored as descriptors for local action features. Mikolajczyk and Uemura (2008) clustered an ensemble of local features to tree representations and related them to action categories. Other approaches (Gupta et al., 2009; Yao & Fei-Fei, 2010) cast action recognition as a two-step structural connectivity task by recognizing parts of objects and understanding actions through pose. Several methods have extended this notion to individual regions (Ikizler-Cinbis & Sclaroff, 2010), poselet clusters (Pishchulin et al., 2013), decision trees (Rahmani et al., 2014), and covariance matrices (Kviatkovsky et al., 2014).

Spatial convolutions. Convolutions can efficiently extract local patterns from visual inputs. An early application of Convolutional Neural Networks (CNNs) to video (Karpathy et al., 2014) temporally fused spatial frame embeddings over pre-defined sets of layers. Others explored the factorization of frame embeddings (Sun et al., 2015), frame ranking (Fernando et al., 2015), pooling (Fernando et al., 2016), salient region focus (Girdhar & Ramanan, 2017; Zong et al., 2021), and relation reasoning between neighboring frames (Zhou et al., 2018a). Le et al. (2011) spatially convolved videos over combinations of the spatial and temporal dimensions. Seminal efforts focused on single volumes to represent motion (Bilen et al., 2016; Chung & Zisserman, 2016; Iosifidis et al., 2012) or learned the correlation and exclusion between action classes (Hoai & Zisserman, 2015). Tran et al. (2018) proposed convolutional blocks based on spatial (2D) and temporal (1D) kernels to create more efficient video models. Lin et al. (2019) reduced redundancies by shifting features at subsequent frames, while later adaptations also included conditional gates (Sudhakaran et al., 2020).

Temporal recursion. A parallel line of research has focused on extracting motion patterns with recurrent layers (Ballas et al., 2015; Dwibedi et al., 2018; Perrett & Damen, 2019; Yue-Hei Ng et al., 2015; Ullah et al., 2017), from the static frame features of spatial CNNs. Several works have jointly encoded frame features and learned changes in appearance over time

with Convolutional LSTMs (Donahue et al., 2015; Srivastava et al., 2015). Similarly, for multi-actor action recognition, Wang et al. (2017b) used three individual pathways with LSTMs for person action, group action, and scene recognition.

Two-stream models. An alternative group of approaches included a parallel motion-specific stream in spatial CNNs. Two-stream models (Simonyan & Zisserman, 2014) encode motion and appearance explicitly with respective optical flow and RGB streams over stacks of frames. Extensions (Feichtenhofer et al., 2016) have fused flow and spatial streams at intermediate layers while other approaches used cross-stream connections (Feichtenhofer et al., 2017), multiple appearance streams (Tu et al., 2018), recurrent layers (Singh et al., 2016), or concatenated appearance and motion volumes (Jain et al., 2015a; Wang et al., 2017c) to share information between the streams. Wang et al. (2016b) used a step-based approach that segmented videos into individual snippets, processed them in parallel, and fused class scores from each snippet. Improvements in inference speeds of two-stream models have been achieved with the addition of motion vectors (Zhang et al., 2016) or key volume mining (Zhu et al., 2016). Although such approaches have established a new research direction in modeling videos, the representation of motion with pre-computed motion features limits the capabilities of learned backbones (Sevilla-Lara et al., 2019).

2.3 Jointly encoding space and time

Time and appearance can also be encoded jointly.

Part-based representations. SpatioTemporal Interest Points (STIPs) (Laptev & Lindeberg, 2003) extended spatial interest point detection methods (Förstner & Gülch, 1987; Harris et al., 1988) to the video domain. Liu and Shah (2008), Oikonomopoulos et al. (2005) explored salient points based on peaks of activity variation. STIP features have been quantized in histograms of codewords (Schuldt et al., 2004). Several approaches have studied action-relevant temporal locations across viewpoints (Yilmaz & Shah, 2006) and view-invariant trajectories (Sheikh et al., 2005). Dollár et al. (2005) proposed modeling periodic motions using sparse distributions of points of interest. This feature extractor prompted subsequent works (Niebles et al., 2008) with actions classified through a codebook of features.

Holistic stochastic representations. Actions have also been modeled based on global information. Efros et al. (2003) created representations for different body parts and regressed towards representations of pre-classified actions. Subsequent works have explored action descriptors focused on object shapes (Gorelick et al., 2006; Jia & Yeung, 2008), movements (Sun et al., 2009), and spatiotemporal salient regions (Wong & Cipolla, 2007). They have also extended existing approaches to multiple features and temporal scales (Amer

& Todorovic, 2012; Liu et al., 2008; Zelnik-Manor & Irani, 2001; Yang et al., 2020b). Later works (Blank et al., 2005) adapted and generalized holistic descriptors (Gorelick et al., 2006) by concatenating 2D silhouettes to form space-time shapes corresponding to action performances. Sadanand and Corso (2012) similarly proposed a bank of volumetrically pooled features containing high-level representations of the actions.

3D CNNs. Orthogonal to hand-crafted features, 2D convolutions have been extended in various ways to 3D spatiotemporal kernels to jointly encode space and time (Baccouche et al., 2011; Ji et al., 2012; Taylor et al., 2010; Tran et al., 2015). Subsequent works have demonstrated the potential of adapting image models to video (Hara et al., 2018), explored video-specific architectures with spatiotemporal volumes across channels (Chen et al., 2018c), and tiled 3D kernels (Hegde et al., 2018). They have also used channel-separated convolutions (Jiang et al., 2019b; Luo & Yuille, 2019; Tran et al., 2019), temporal residual connections (Qiu et al., 2017), global feature fusion (Qiu et al., 2019), resolution reduction (Chen et al., 2019; Stergiou & Poppe, 2021b), and related appearance to spatiotemporal embeddings (Wang et al., 2018c; Zhou et al., 2018d). Carreira and Zisserman (2017) integrated 3D convolutions into two-stream models for motion-implicit appearance representations in the RGB stream and motion-explicit representations in the optical flow stream. Several works have focused on improving the efficiency of action recognition architectures (Feichtenhofer, 2020; Kondratyuk et al., 2021; Liu et al., 2022i). They have used visual context from the scenes of actions, by either scene-type objectives (Choi et al., 2019), decoupling scene and motion features (Wang et al., 2021a), multi-domain information concatenation (Kapidis et al., 2023), or by fusing motion and scene information (Stergiou & Poppe, 2021a). To better extract temporal information, Feichtenhofer et al. (2019) proposed a dual pathway video model with a slow pathway operating over low frame rates for spatial semantics and a fast pathway with a high frame rate for motion. Similarly, Wang et al. (2020a) included a contrastive objective for learning the pace in videos. Xu et al. (2019a) explored temporal reasoning by including clip order prediction as an additional task to improve action recognition. The extension of 3D CNNs to longer sequences by segmenting videos with multiple temporal patches has also been attempted (Ji et al., 2020; Hussein et al., 2019; Varol et al., 2017).

Spatiotemporal attention. Attention is an effective approach for learning feature correspondences over space and time. Sharma et al. (2015) used visual attention to localize action regions from CNN features with recurrent layers. Du et al. (2017) attended over spatial features across multiple frames based on their relevance to the action. Similarly, Chen et al. (2018b) aggregated and propagated global information by attending over convolution features. Wang et al. (2018e)

introduced non-local operations with bi-directional attention blocks over convolutions. Another early application of attention (Girdhar et al., 2019) was based on region proposals and the creation of feature banks (Wu et al., 2019a) in longer videos. The introduction of Vision Transformers (ViTs) (Dosovitskiy et al., 2020) that encode visual information through region-based tokenization led to video-based adaptations that explored different spatiotemporal attention configurations (Arnab et al., 2021a; Bertasius et al., 2021). Others have explored token selection (Bulat et al., 2021; Ryoo et al., 2021; Zha et al., 2021), and the inclusion of contextual information (Kim et al., 2021c). Liu et al. (2022j) introduced shifted non-overlapping attention windows to share information across patches. Lu et al. (2024b) additionally used attention over motion-aligned input volumes. Feature hierarchies and latent resolution reductions have led to more compute- (Fan et al., 2021; Li et al., 2022e) and memory-efficient (Wu et al., 2022b) architectures. Recent models such as MViT (Yan et al., 2022), Hiera (Ryali et al., 2023), UniFormer (Li et al., 2022c), and MooG (van Steenkiste et al., 2024), improved both the performance and capacity of video models. SSL has also shown great promise with pretext tasks based on contrastive learning (Chen et al., 2020c) or token masking (He et al., 2022b). Xing et al. (2023) increased the complexity of the contrastive objective with pseudo labels and token mixing from different inputs. Masked autoencoders have also been extended to video data (Feichtenhofer et al., 2022; Wei et al., 2022a). Subsequent works have explored adaptive token masking (Bandara et al., 2023), double masking on both the encoder and decoder (Wang et al., 2023d), token fusion (Kim et al., 2024b), and teacher-student masked autoencoders (Wang et al., 2023f).

Video-language models. Recently, language semantics from Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023) have been used as a supervisory signal for vision tasks (Li et al., 2023a; Liu et al., 2024a; Radford et al., 2021). Initial efforts (Zellers et al., 2021) matched frame-level encodings to corresponding LLM embeddings of captions. Other approaches have optimized image-based encodings over frames by pooling spatial tokens (Yu et al., 2022a), including cross-modal skip connections (Xu et al., 2023b), cross-attending modalities (Alayrac et al., 2022), and jointly attending visual and text embeddings (Maaz et al., 2023). As static features provide only an appearance-based view, works have also used spatiotemporal Vision-Language models (VLMs) (Piergiovanni et al., 2024) and extended the training objective (Lu et al., 2024a; Zhao et al., 2024b) to a two-step SSL pre-training with video-to-text alignment and video masking.

3 Video datasets comprising human actions

Significant efforts have been made to collect video datasets for various action understanding tasks. We discuss the main challenges associated with dataset collection in Section 3.1. We then explore two broad dataset types based on target tasks and use cases. The first set includes general-purpose datasets for pre-training and model evaluation. The second set of datasets has been collected to evaluate models on specific modalities or domains. The sets are discussed in Section 3.2 and Section 3.3, respectively.

3.1 Data collection challenges

Ensuring sufficient diversity of people, scenarios, and activities is important when amassing video datasets at scale. Although in recent years scalability has been achieved by transiting from locally-sourced datasets (Caba Heilbron et al., 2015; Laptev et al., 2008; Shahroudy et al., 2016; Schudt et al., 2004; Soomro et al., 2012) to multi-team-multi-year efforts (Grauman et al., 2022; Li et al., 2024e; Yang et al., 2024a) dataset diversity remains subjective which can result to semantically overlapping (Khamis & Davis, 2015; Wray & Damen, 2019) or ambiguous (Kim et al., 2022; Sigurdsson et al., 2017) labels. As video understanding is a multifaceted topic relating to vision, robotics, and augmented reality, collecting **meaningful scenarios and annotations** also presents difficulties. Several datasets have used surveys (Caba Heilbron et al., 2015; Grauman et al., 2022, 2024; Lin et al., 2023a) or relied on meta-data from online videos (Chen et al., 2023a; Miech et al., 2019; Smaira et al., 2020; Yang et al., 2024a) as guides for collection. Despite such efforts, sourcing is still bound to preliminary data assumptions (Rahaman et al., 2022), inconsistencies across annotations (Moltisanti et al., 2017), and difficulties in task definitions (Alwassel et al., 2018). The collection pipelines also require scalability. In recency, **greater automation** in video collection has been achieved with the use of embeddings from vision encoders (Chen et al., 2020a; Huang et al., 2024e; Zhu et al., 2024a), and LLM-generated descriptions (Fu et al., 2024; Li et al., 2024g; Mangalam et al., 2023).

3.2 General datasets

The past two decades have seen a significant increase in dataset size, leading to more robust baselines. We present widely-adopted benchmarks chronologically in Table 2. The primary focus of initial benchmarks (Schudt et al., 2004; Gorelick et al., 2007) has been the categorization of simple actions such as *walking* and *hand waving*. Subsequent datasets predominantly comprised videos from either TV shows/movies (Laptev & Pérez, 2007; Laptev et al., 2008; Marszalek et al., 2009; Patron-Perez et al., 2010; Kuehne

Table 2 Action understanding datasets. Works are grouped by year of release (Y). The number of classes, video instances, and actors are denoted with #Cls, Inst, and Act. The average duration per annotation is denoted as AD. Short descriptions per dataset appear in the Context column

Y	Dataset	#Cls/Inst/Act/AD	Context
2004-2007	KTH (Schuldt et al., 2004)	6/2K/25/2.5s	Grayscaled videos of motions
	Weizmann (Gorelick et al., 2007)	10/90/8/12s	Low-res. atomic motions
	Coffee&Cigarettes (Laptev & Pérez, 2007)	2/245/5/5s	Smoking/drinking in movies
	CASIA Action (Wang et al., 2007)	15/1446/24/NA	Outdoor activities
2008-2014	UCF Sports (Rodriguez et al., 2008)	9/150/<100/5s	Sports videos
	Hollywood (Laptev et al., 2008)	8/475/<100/16s	Actions in movies
	UT-interaction (Ryoo & Aggarwal, 2009)	6/90/60/17s	Dyadic human interactions
	CMU-MMAC (la Torre Frade et al., 2008)	5/182/43/7m	Multi-view recipe preparations
	UCF-11 (Liu et al., 2009)	11/1K/100+/5s	Actions in YouTube videos
	Hollywood2 (Marszalek et al., 2009)	12/3K/100+/12s	Actions from movies
	TV-HI (Patron-Perez et al., 2010)	4/300/100+/3s	Interactions in TV shows
	UCF-50 (Reddy & Shah, 2013)	50/5K/100+/15s	Web-sourced videos
	Olympic Sports (Niebles et al., 2010)	16/800/100+/3s	Actions in sports
	HMDB-51 (Kuehne et al., 2011)	51/7K/100+/3s	Actions from movies
	CCV (Jiang et al., 2011)	20/9K/100+/80s	Web-sourced videos
	UCF-101 (Soomro et al., 2012)	101/13K/100+/15s	Action with hierarchies
	CAD-60 (Sung et al., 2012)	12/60/<30/45s	Atomic actions in RGB-D
	MPII (Rohrbach et al., 2012)	65/5.6K/100+/11m	Web-source actions
	ADL (Pirsiavash & Ramanan, 2012)	32/436/20/1.3s	Videos of daily activities
	50 Salads (Stein & McKenna, 2013)	17/899/25/37s	Salad making videos
	J-HMDB (Jhuang et al., 2013)	21/928/100+/1.2s	Videos with joints positions
	CAD-120 (Koppula et al., 2013)	12/120/<60/45s	Extension of CAD-60
	Penn Action (Zhang et al., 2013)	15/2.3K/100+/2s	Web-sourced atomic actions
	Sports-1M (Karpathy et al., 2014)	487/1M/1,000+/9s	Sports actions/activities
2015-2018	EGTEA Gaze+ (Li et al., 2015)	106/15K/32/28s	Egocentric actions w/ gaze
	ActivityNet-100 (Caba Heilbron et al., 2015)	100/5K/100+/2m	Untrimmed web videos
	Watch-n-Patch (Wu et al., 2015)	21/2K/7/30s.	Daily activities in RGB-D
	NTU-RGB-60 (Shahroudy et al., 2016)	60/57K/40/2s.	Multi-sensory actions
	ActivityNet-200 (Caba Heilbron et al., 2015)	200/15K/100+/2m	ActivityNet-100 extension
	YouTube-8M (Abu-El-Haija et al., 2016)	NA/8M/NA/NA	Multi-labelled videos
	Charades (Sigurdsson et al., 2016)	157/67K/267/30s	Daily activities videos
	ShakeFive2 (Van Gemeren et al., 2016)	5/153/33/7s	Interactions with pose data
	DALY (Weinzaepfel et al., 2016)	10/510/100+/4m	Untrimmed YouTube videos
	OA (Li & Fritz, 2016)	48/480/<100/5s	Ongoing actions
	CONVERSE (Edwards et al., 2016)	10/NA/NA/NA	Human interactions
	TV-Series (De Geest et al., 2016)	30/6,2K/100+/2s	Actions from TV series
	Volleyball (Ibrahim et al., 2016)	6/1.4K/<100/<1s	Group actions in volleyball
	MSR-VTT (Xu et al., 2016)	200K/7.1K/1,000+/20s	Video captions
	Okutama Action (Barekatin et al., 2017)	12/4.7K/~400/60s	Aerial views of action
	K-400 (Kay et al., 2017)	400/306K/1,000+/10s	Web-sourced short actions
	Smthng-Smthng v1 (Goyal et al., 2017a)	174/109K/100+/4s	Human actions with objects
	MultiTHUMOS (Yeung et al., 2018)	65/39K/100+/3s	Densely labeled actions
	Diving-48 (Li et al., 2018b)	48/18K/NA/3s	Diving sequences

Table 2 continued

Y	Dataset	#Cls/Inst/Act/AD	Context
2019-now	EK-55 (Damen et al., 2018)	2,747/40K/35/3s	Egocentric actions in kitchens
	K-600 (Carreira et al., 2018)	600/495K/100+/10s	Extension of K-400
	VLOG (Fouhey et al., 2018)	30/122K/10.7K/10s	Actions in lifestyle VLOGs
	AVA (Gu et al., 2018)	80/430/100+/15m	Localized atomic actions
	NTU-RGB-120 (Shahroudy et al., 2016)	120/114K/106/2s.	Multi-sensory actions
	Charades-Ego (Sigurdsson et al., 2018)	156/7.8K/100+/9s	Daily indoor activities
	Smthng-Smthng v2 (Goyal et al., 2017a)	174/221K/100+/4s	Human actions with objects
	K-700 (Carreira et al., 2019)	700/650K/1,000+/10s	Extension of K-600
	Moments in Time (Monfort et al., 2019)	339/1M/1,000+/3s	Short dynamic scenes
	HACS (Clips) (Zhao et al., 2019)	200/1.5M/1,000+/2s	Action over fixed durations
	IG65M (Ghadiyaram et al., 2019)	NA/65M/NA/NA	Actions in Instagram videos
	Toyota Smarthome (Dai et al., 2022a)	31/16K/18/21m	Senior home activities
	AViD (Piergiovanni & Ryoo, 2020)	887/450K/1,000+/9s	Anonymized videos
	HVU (Diba et al., 2020)	3K/572K/1,000+/10s	Hierarchy of semantics
	Action-Genome (Ji et al., 2020)	453/10K/100+/1s	Daily home activities
	K-700 (2020) (Smaira et al., 2020)	700/647K/1,000+/10s	Update of K-700
	FineGym (Shao et al., 2020)	530/33K/100+/10m	Gymnastics videos
	RareAct (Miech et al., 2020a)	122/7.6K/100+/10s	Unusual actions
	HAA500 (Chung et al., 2021)	500/10K/1,000+/2s	Atomic actions
	MultSports (Li et al., 2021c)	4/3.2K/100+/21s	Localized sports actions
	MOMA (Luo et al., 2021)	136/12K/100+/10s	Hierarchical actions
	WebVid-2M (Bain et al., 2021)	NA/2M/1,000+/4s	Video-image pairs
	HOMAGE (Rai et al., 2021)	453/5.7K/40/2s	Extension of (Ji et al., 2020)
	EK-100 (Damen et al., 2022)	4,053/90K/37/3s	Egocentric actions
	FineAction (Liu et al., 2022g)	106/103K/1,000+/7s	Hierarchies for TAL
	EGO4D (Grauman et al., 2022)	1000+/9.6K/931/48s	Diverse egocentric videos
	Assembly-101 (Sener et al., 2022)	1.3K/4.3K/53/2s	Procedural activities
	Ego-Exo-4D (Grauman et al., 2024)	689/5,035/740/5m	Multimodal multi-view videos

et al., 2011) or sports footage (Rodriguez et al., 2008; Liu et al., 2009; Reddy & Shah, 2013; Niebles et al., 2010). Important steps towards establishing large-scale datasets for the video domain were made with the introduction of Sports-1M (Karpathy et al., 2014), YouTube-8M (Abu-El-Haija et al., 2016), and Kinetics (Carreira & Zisserman, 2017) that include web-sourced videos of a diverse range of actions. Evident from their sizes shown in Figure 3, these datasets paved the way as general benchmarks for models that can subsequently be adapted to smaller, more niche datasets such as UCF-101 (Soomro et al., 2012) and ActivityNet (Caba Heilbron et al., 2015). Despite their size and use in multiple downstream tasks, there is still room to address specific action understanding tasks or modalities supplementary to vision. Domains such as egocentric vision, human-object interaction recognition, and hierarchical action understanding have gained popularity, prompting the creation of domain-specific datasets. EGTEA Gaze+ (Li et al.,

2015), EPIC KITCHENS (Damen et al., 2022), and later EGO4D (Grauman et al., 2022) have been the main benchmarks for egocentric vision. Something-Something (Goyal et al., 2017a) and Charades (Sigurdsson et al., 2016) have been predominantly used as benchmarks for object-based actions with a greater focus on temporal information. Datasets such as Diving-48 (Li et al., 2018b) and FineGym (Shao et al., 2020) incorporate semantic hierarchies in their annotations. More recent datasets have focused on tasks related to action recognition including instruction learning (Alayrac et al., 2016; Bansal et al., 2022; Ben-Shabat et al., 2021; Liu et al., 2024h; Ohkawa et al., 2023; Sener et al., 2022; Tang et al., 2019), action phase alignment (Sermanet et al., 2017), repeating action counting (Dwibedi et al., 2020, 2024; Hu et al., 2022a; Runia et al., 2018; Zhang et al., 2020a), action completion prediction (Epstein et al., 2020), driver behavior recognition (Martin et al., 2019; Ortega et al., 2020), anomaly detection (Acsintoae et al., 2022; Liu et al., 2018b; Lu et

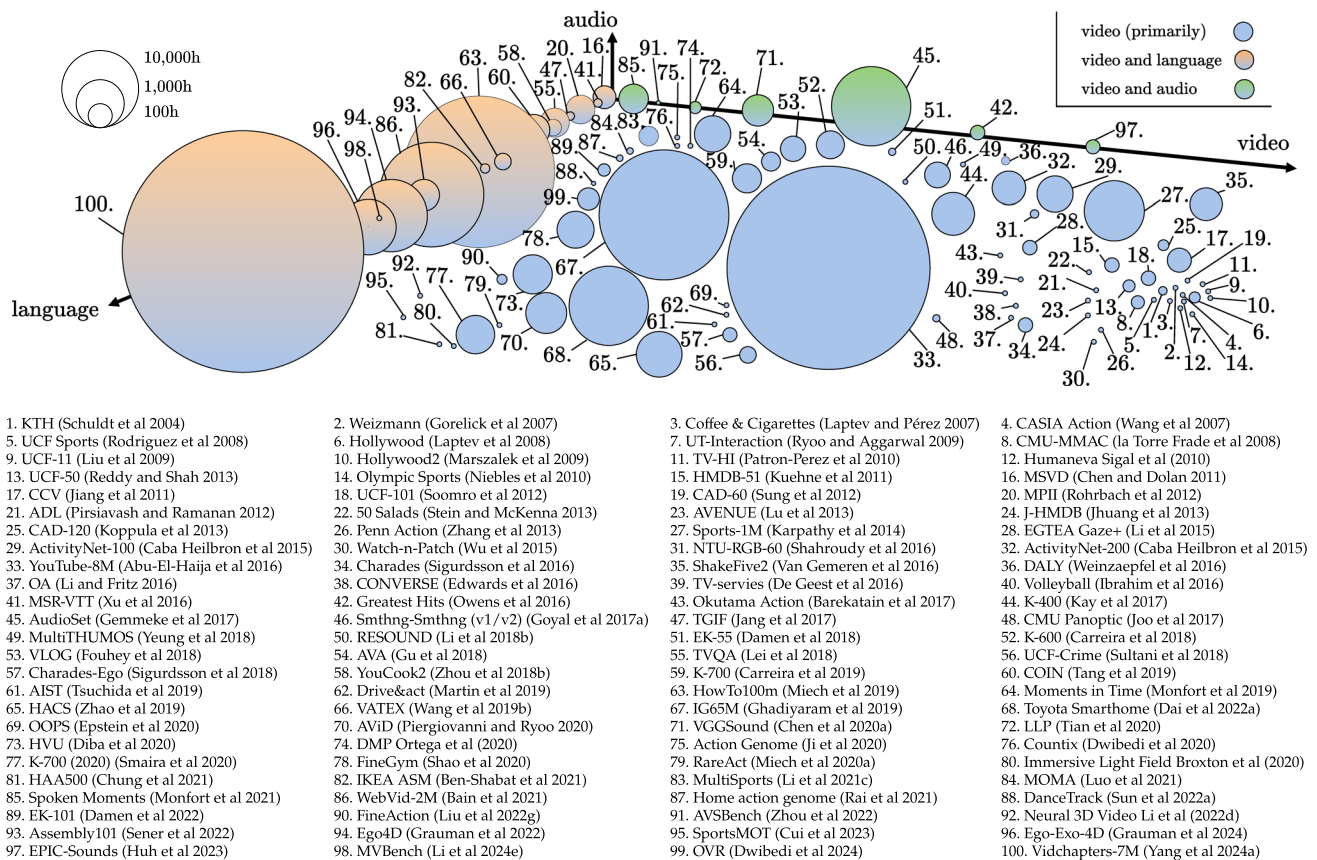


Fig. 3 Datasets compared by total dataset duration and primary modality. Circle sizes correspond to the (approximate) summed duration of all videos in the datasets. Recent datasets (i.e., > 80) have longer total running times and include additional modalities such as language or audio

al., 2013; Sultani et al., 2018; Wu et al., 2020a), hand-object interactions (Chao et al., 2021; Garcia-Hernando et al., 2018; Hampali et al., 2020; Kwon et al., 2021; Moon et al., 2020; Mueller et al., 2017), and object state change detection in actions (Souček et al., 2022).

3.3 Domain- and modality-specific datasets

Apart from general-purpose datasets, several benchmarks have been designed to evaluate model capabilities of specific aspects of action understanding. We overview of benchmarks in three groups: based on the holistic understanding of scenes from multiple viewpoints, and with supplementary modalities such as language and audio.

Multi-view. Initial efforts to compile multi-view videos included a small number of subjects (Sigal et al., 2010) or synthetic data (Ionescu et al., 2013). High-quality multi-view videos depend highly on the hardware and setup (Wang et al., 2023e). CMU panoptic (Joo et al., 2017) captured group interactions within a dome with 480 cameras. Interactions included social settings, games, dancing, and musical performances. ZJU-Mocap (Peng et al., 2021) comprised dynamic videos of human motions from 20 cameras. The Immersive

Light Field dataset (Broxton et al., 2020) contains videos with 6 degrees of freedom from a camera rig consisting of 46 action cameras. Multi-view datasets are collected for a variety of target tasks, including dance sequence reconstruction (Tsuchida et al., 2019), and the dynamic synthesis of indoor spaces in which actions take place (Dai et al., 2022a; Tschernetzki et al., 2024).

3D vision. 3D human action reconstruction initially relied on synthetic data (Anguelov et al., 2005; Bronstein et al., 2010) or lacked detailed ground-truth annotations (Koppula et al., 2013; Li et al., 2010; Shahroudy et al., 2016; Sung et al., 2012). (D)FAUST (Bogo et al., 2014, 2017) introduced a full pipeline for capturing high-resolution deformations. The collected 3D models of 300 meshes from static scans later increased to 40K meshes from dynamic scans. With the same multi-camera setting, DYNA (Pons-Moll et al., 2015) collected meshes by aligning 3D scans to template meshes using only geometric information. Other 3D data collection efforts have been specific to action synthesis in indoor (Li et al., 2022d) and outdoor (Lin et al., 2021b; Yoon et al., 2020) settings, depth-based procedural learning (Ben-Shabat et al., 2021; Sener et al., 2022), human-object interaction tracking (Bhatnagar et al., 2022), novel view synthesis (Jiang et al.,

2022; Peng et al., 2021), language captioning for poses (Delmas et al., 2022), or representing human-relevant attributes such as clothes and hair dynamics (Black et al., 2023). Large-scale collections that unify existing datasets also exist. Guo et al. (2020) created a dataset for 3D human motions by revamping Joo et al. (2017); Shahroudy et al. (2016); Zou et al. (2020). Similarly, Mahmood et al. (2019) combined a corpus of 15 archival marker-based mocap datasets while Lin et al. (2023b) created a superset of datasets by sourcing videos (Cai et al., 2022b; Chung et al., 2021; Taheri et al., 2020; Tsuchida et al., 2019; Zhang et al., 2022c) relevant to LLM motion text prompts. Recently, space-time-depth datasets and benchmarks have also been introduced for ego-centric data for tracking human-object interactions (Liu et al., 2022f; Perrett et al., 2025), mistake detection in procedural tasks (Wang et al., 2023h), multi-task augmented reality (Grauman et al., 2024; Lv et al., 2024; Pan et al., 2023), and surface estimation and reconstruction (Straub et al., 2024).

Video-language. In recent years, language has been integrated into vision methods as a natural extension to represent high-level semantics. Commonly, learning to map textual concepts and visual representations in a shared embedding space has been a widely adopted strategy by many video tasks (Amrani et al., 2021; Gabeur et al., 2020; Liu et al., 2019; Miech et al., 2020b). Initial video-language datasets (Chen & Dolan, 2011; Xu et al., 2016) were based on short video snippets and short textual descriptions of actions. More recent efforts also provide multilingual descriptions (Wang et al., 2019b). Video question-answering is a popular language-based task (Jang et al., 2017; Lei et al., 2018; Li et al., 2024e; Oncescu et al., 2021; Rawal et al., 2024; Xiao et al., 2021). The order of instructions has been of great interest in longer videos since the introduction of HowTo100M (Miech et al., 2019) and YouCook2 (Zhou et al., 2018b). Benchmarks have also been proposed for other long-form tasks such as moment retrieval (Rohrbach et al., 2015; Song et al., 2024; Yang et al., 2024a), frame extraction (Li et al., 2024a), multimodal open-ended question answering (Fu et al., 2024; Ying et al., 2024), and long-term reasoning (Chandrasegaran et al., 2024; Fei et al., 2024b; Mangalam et al., 2023).

Audio and vision. Human perception often relies on the inclusion of audio for understanding actions, especially in conditions where appearance may lead to ambiguous predictions. Audioset (Gemmeke et al., 2017) is the largest audio-visual action dataset containing 2.1M clips across a long-tail distribution of 527 classes. VGG-Sound (Chen et al., 2020a) is another common benchmark with a uniform distribution of 200K videos over 300 classes. Datasets have also been collected for specific tasks such as audio-visual semantic segmentation (Zhou et al., 2022), audio-visual video parsing (Tian et al., 2020), material sound and action classification (Huh et al., 2023; Owens et al., 2016), and video captioning (Monfort et al., 2021).

The introduction of both general and task-specific datasets has improved the exploration of video tasks and standardized evaluation protocols. Main lines of research and challenges of these tasks are overviewed next.

4 Recognizing observed actions

The recognition of actions in videos is a fundamental computer vision research theme. Relevant tasks focus on different aspects of the actions observed in full. We start by discussing approaches for optimizing model inputs in Section 4.1. We then overview popular temporal-based recognition tasks in Section 4.2. Tasks based on the semantic relationships between language and video are discussed in Section 4.3, whereas audio-visual and other multimodal approaches appear in Section 4.4.

4.1 Video reduction methods

Video inputs typically consist of tens to hundreds of highly visually similar frames. The uniform use of all frames can lead to an unsustainable computational burden. However, humans process stimuli selectively (Eagleman, 2010). Several recognition approaches, as shown in Figure 4, aim to reduce compute and improve memory utilization by considering inputs selectively.

4.1.1 Challenges

Reducing frame-level redundancies in videos requires a high-level understanding of each temporal segment's relevance. The distinction and selection of relevant segments directly impact information loss. Long and complex scenes present significant challenges to video reduction methods. This *uneven context inclusion* requires more efficient utilization of the model's capacity.

4.1.2 Approaches

Among the most common approaches for reducing redundancies is *frame sampling*. Works on frame sampling rely on policy networks that select frames based on the action's complexity (Ghodrati et al., 2021; Yeung et al., 2016), video context correspondence (Wu et al., 2019c), or changes in the target class' probability (Korbar et al., 2019). Wang et al. (2021b) used a recurrent network to localize action-relevant regions. Subsequent extensions targeted early stopping (Wang et al., 2022d) and related local and global features to determine action-relevant patches (Wang et al., 2022e). Xia et al. (2022b) used pseudo labels obtained by computing the embedding distance to class centroids to distinguish individual frames as salient and non-salient. Other approaches have

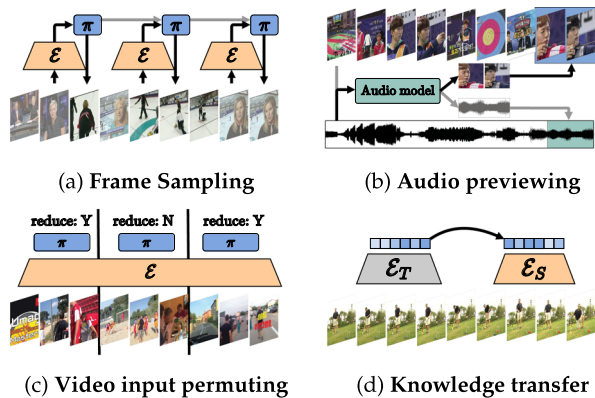


Fig. 4 Redundancy reduction methods include (a) selection of task-specific salient frames, (b) use of supplementary modalities such as audio to preview relevant regions to sample from, (c) input permutations to compress irrelevant frames and segments, and (d) using embeddings from a teacher model as targets

used reward functions based on predictions from the selected frames (Wu et al., 2020d), combined frame-level and video-level predictions (Gowda et al., 2021), optimized towards balancing accuracy and number of frames used (Wu et al., 2019b), or removed tokens in transformer architectures (Wu et al., 2024b).

A related set of approaches has extended unimodal frame sampling with *audio previewing*. Gao et al. (2020) used both frame and audio features with a recurrent network to predict the next informative moment in the video. The video resolution used by the model was determined based on discovered informative parts in the audio stream. Similarly, Nugroho et al. (2023) used a saliency loss to localize the informative audio segments from which the corresponding video frames can be sampled.

Although coarse frame sampling can be beneficial in short videos, selecting a limited number of frames in longer videos with a broader context can result in information loss. Another line of research thus studies redundancy reduction through *video input permutations*. These works change frame resolutions based on classifier confidence (Meng et al., 2020) or quantize frames at different precision (Abati et al., 2023; Sun et al., 2021b). Zhang et al. (2022d) used a two-branch approach for lightweight computations over large, less-relevant segments, and assigned more compute for segments with relevant context similar to (Feichtenhofer et al., 2019).

Recent efforts have also used *knowledge distillation* to improve the training efficiency of video model pipelines. Ma et al. (2022a) reduced computations by learning to match student network features from videos with reduced resolution to the full-resolution features from a teacher network. Kim et al. (2021b) extended this approach by using cross-attention to learn the correspondence between teacher and

student features. Distillation approaches have also used non-vision teacher models. Lei et al. (2021b) bound language embeddings to sparsely sampled clips from long videos while Xia et al. (2022a) used embeddings from textual event-object relations to discover salient frames. Tan et al. (2023b) proposed a reconstruction approach for interpolating egocentric video features using embeddings from partial frames and the camera motion for the unobserved frames.

4.1.3 Future outlooks

Context-aware models can significantly improve both processing times and performance. Although most video reduction methods are primarily evaluated on classification or detection, more recent action understanding models have been optimized on multi-task and multi-domain objectives. This makes the discovery of relevant frames more difficult. For example, suppressing background frames is sub-optimal for episodic memory in which specific locations or attributes of objects not directly relevant to a current action need to be inferred. We expect that future redundancy reduction methods will focus more on preserving general scene information rather than ensuring that semantics of objects in the scene are not lost. This can be potentially achieved by discovering informative frames for diverse tasks or distilling scene context in low-memory representations, e.g. language embeddings.

4.2 Temporal-based tasks

The perception of actions across time is a complex capability of human cognition. Understanding the timing of events is crucial for developing motor memory (Eagleman, 2010) for actions such as moving, speaking, and determining the causality of perceived temporal patterns. The importance of processing temporal information efficiently by computer vision systems has been shown through both standard performance metrics and semantic benchmarks (Albanie et al., 2020; Stergiou & Deligiannis, 2023). We provide a visualization of temporal-based tasks in Figure 5 with the main challenges and task details discussed below.

4.2.1 Challenges

For most tasks, action categories are inferred directly without an explicit notion of their complexity based on **levels of abstraction**, e.g., atomic movements, composite motions, singular actions, or general activities. Although some datasets include action hierarchies (Shao et al., 2020; Li et al., 2018b), these relationships are only used by a handful of existing works (Long et al., 2020). Different levels of abstraction typically have different temporal ranges, which require different approaches to process visual inputs. Moreover, there

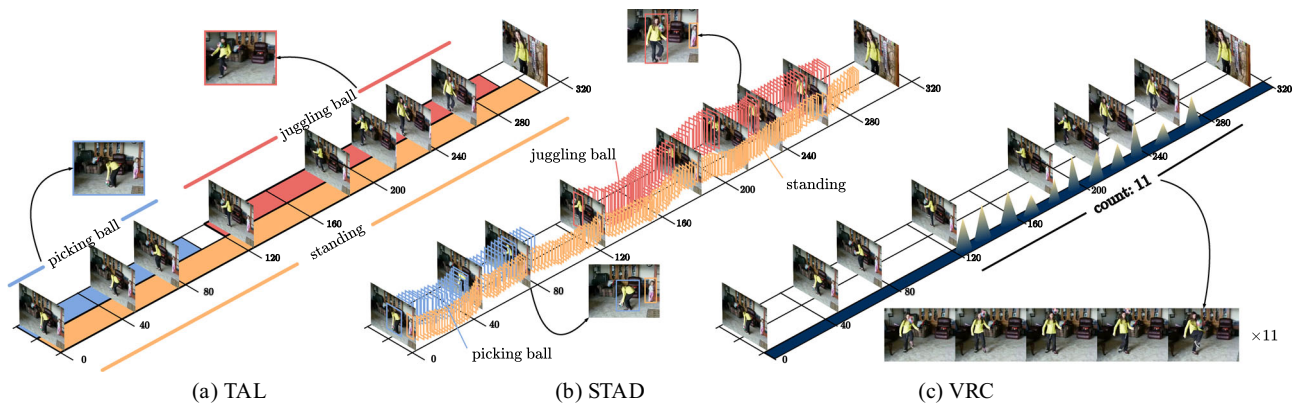


Fig. 5 Visualization of temporal-based tasks. (a) Temporal Action Localization (TAL) discovers the start and end times of individual actions. In contrast, (b) Spatio-Temporal Action Detection (STAD) is more complex as it requires temporally and spatially localizing actions

with bounding boxes for actors and objects over time. Distinctively, (c) Video Repetition Counting (VRC) is not based on action labels and instead requires counting repetitions of actions or motions in an open-set setting. Video source from Kay et al. (2017)

can be substantial temporal variations across class instances. This leads to larger temporal distances between discriminative information in videos, requiring proper extraction and modeling of long-range dependencies. We discuss solutions to cope with these variations in this section.

4.2.2 Temporal localization

A well-established video task is the discovery and classification of the actions performed alongside their temporal segments. Temporal Action Localization (TAL) aims to infer the action categories alongside the start and end times of the corresponding locations in untrimmed videos.

Early attempts have used Improved Dense Trajectories (Wang & Schmid, 2013) and Fisher vectors (Oneata et al., 2013) to model the temporal dynamics of local points in scenes. Shou et al. (2016) was one of the first to approach TAL with a joint action proposal and classification objective with a regional CNN. This joint optimization has been adapted with spatial- and temporal-only networks (Lin et al., 2018; Paul et al., 2018; Wang et al., 2017a), regional proposal selection (Chao et al., 2018; Xu et al., 2017b), and intra-proposal relationships with graph convolutions (Zeng et al., 2019) in subsequent works. Shou et al. (2017) predicted granularities at a frame level by transposing the temporal resolution of pretrained video encoders. More recent approaches for TAL can be categorized into three broad categories.

One-stage. Most similar to the aforementioned methods, one-stage approaches localize and classify actions in a single step by using hierarchical embeddings from feature pyramids (Lin et al., 2021a; Liu & Wang, 2020; Shi et al., 2023; Zhang et al., 2022b) or relating relevant video segments (Shou et al., 2018; Yang et al., 2020c). Recently, Yan et al. (2023b) integrated scene semantic context for TAL with the inclu-

sion of vision-language encoders. The start and end frames of an action can often be ambiguous. A set of approaches have relaxed deterministic start-end times objectives either by weighing the training loss with the importance of each frame (Shao et al., 2023) or by learning distributions of possible start and end times (Moltisanti et al., 2019). These approaches aimed to introduce a variance prior to the typical duration of the action.

Two-stage. Two-stage approaches disentangle the optimization into separate parts. Zhai et al. (2020) combined proposals from spatial- and temporal-only streams into a fused final prediction. Chen et al. (2022a) used long- and short-range temporal information to refine the confidence of the generated action proposals. Huang et al. (2019) decoupled classification and localization to two objectives during training with two separate models that use cross-modal connections for exchanging information. Some works have focused on maximizing the embedding difference between representations of frames from action segments and non-relevant frames, either with positive and negative instances (Luo et al., 2020; Zhang et al., 2021a) or by a scoring function (Rizve et al., 2023). Methods have also improved the features of backbone encoders with TAL-relevant pretext tasks (Zhang et al., 2022a). Several two-stage methods process videos holistically (Alwassel et al., 2021; He et al., 2022a; Liu et al., 2021f; Qing et al., 2021). Alwassel et al. (2021) encoded local features over a sliding window using aggregated features from multiple local encoders. The model was optimized with a dual objective for discovering action regions and classifying every segment. Graphs have also been adopted for TAL with Bai et al. (2020) using generated candidate proposals for the graph's start and end edges and their connected nodes. Zhao et al. (2021) created a graph of hierarchical features over multiple temporal resolutions. More recent approaches

(Nag et al., 2023) have formulated proposal prediction as a denoising task with noisy action proposals as input to a diffusion model conditioned on the video. Recent approaches have also used pretrained LLMs for TAL with specific Contrastive Language-Image pretraining (CLIP) query embeddings (Ju et al., 2022), video representation masks adapted to the CLIP text encoder space (Nag et al., 2022), and CLIP text and visual features that are correlated to foreground masks learned from the video (Phan et al., 2024).

Detection Transformers (DETR). A recent set of methods is based on adapting image-based DETR (Carion et al., 2020) to TAL. These approaches rely on transformer encoder-decoders to create regional proposals optimized with bipartite matching. Tan et al. (2021b) adapted DETR to video with a matching scheme of multiple positive action proposals to address sparsity in temporal annotations. Subsequent works have optimized the detection pipeline by either including dense residual connections (Zhao et al., 2023) or caching short-term features (Cheng & Bertasius, 2022; Hong et al., 2022a). Liu et al. (2024e) increased the model capacity by training intermediate adapters propagating information to the decoder from intermediate frozen encoder layers. Approaches have also explored training recipes with sparsely updating model layers (Cheng et al., 2022), vision-language pretraining distillation (Ju et al., 2023), proposal hierarchies (Wu et al., 2023a), and end-to-end TAL encoder-decoder optimization (Liu et al., 2022e). Works aiming to reduce training requirements have also been based on language embeddings with tuned visual-language projectors (Liberatori et al., 2024) and by start/end queries (Aklilu et al., 2024).

4.2.3 Spatiotemporal detection

SpatioTemporal Action Detection (STAD) is related to TAL but aims to jointly localize actions temporally and spatially detect action-relevant actors and objects. The main challenge of STAD methods is consistently linking detections and temporal action proposals across frames. Similar to TAL, two general directions can be used to overview relevant literature.

Two stages. Building upon the advancements of image-based object detectors (Girshick et al., 2014; Girshick, 2015), the majority of STAD approaches first detect objects and then temporally localize actions by tracking object candidates (Jain et al., 2014; Weinzaepfel et al., 2015), ROI-pooling RGB and flow features (Peng & Schmid, 2016), refining proposals iteratively (Soomro et al., 2015), aligning source and target domain features (Agarwal et al., 2020), or using the general action level in the video as context (Mettes et al., 2016). Li et al. (2018a) built upon prior two-stage detection works and incorporated recurrent proposals to include temporal context. Other approaches that focus on

temporal information Singh et al. (2017) used the arrow of time with different portions of the video detected at each step. Later benchmarks included longer videos to focus on activity-related tasks (Gu et al., 2018), enabling the greater exploration of context with feature banks (Feng et al., 2021b; Pan et al., 2021a; Tang et al., 2020a; Wang & Gupta, 2018; Wu et al., 2019a, 2022b) and supplementary object information (Arnab et al., 2021b; Hou et al., 2017; Zhang et al., 2019d). Additional information such as keyframe saliency maps (Li et al., 2020b; Ulutan et al., 2020), hands and poses (Faure et al., 2023), actor-object relations (Sun et al., 2018), and SSL (Wang et al., 2023d) have also been explored. Alwassel et al. (2018) analyzed the benefits of two-stage approaches and showed that they are primarily performant in handling temporal context. However, they also note that a significant limitation of two-stage approaches is that features are computed from backbones trained over auxiliary video tasks, potentially missing specific discriminative information.

Single-stage. Drawing inspiration from single-stage object detection methods (Carion et al., 2020; Redmon et al., 2016; Liu et al., 2016), single-stage STAD approaches use an end-to-end trained unified framework for joint localization and detection (Chen et al., 2021b; Girdhar et al., 2019; Zhu et al., 2024c). Ntinou et al. (2024) extended the bipartite matching loss from Carion et al. (2020) to spatio-temporal tokens. Other approaches used adaptive feature sampling (Wu et al., 2023d), conditionally modeled visual features based on motion (Zhao & Snoek, 2019), and contrasted different views (Kumar & Rawat, 2022). Directly predicting tubelets has also been adopted by recent approaches (Gritsenko et al., 2024; Kalogeiton et al., 2017; Song et al., 2019; Yang et al., 2019; Zhao et al., 2022). Kalogeiton et al. (2017) stacked embeddings from a backbone applied over a sliding window and regressed both classes and tubelets over the entire video. Zhao et al. (2022) used an encoder-decoder to generate tubelet queries and cross-attended them to visual features. Gritsenko et al. (2024) generated candidate tubelets from condensed query representations cross-attended by features from each frame. Beyond STAD, tubelets have also been used as a self-similarity pretraining objective (Thoker et al., 2023) to enforce correspondence of videos from different domains but with similar local motions.

4.2.4 Repetition counting

Video Repetition Counting (VRC) aims to count the number of action repetitions. In contrast to TAL and STAD, VRC is an open-set task and does not require action categories.

Early works on signal periodicity (Thangali & Sclaroff, 2005) have decomposed signal repetition with a Fourier analysis (Albu et al., 2008; Briassouli & Ahuja, 2007; Azy & Ahuja, 2008; Cutler & Davis, 2000; Pogalin et al., 2008).

Table 3 Self-Supervised Learning (SSL) methods and video task adaptations. Three learning paradigms are used to group pretext tasks

Learning	Task	Method	Video adaptations ¹
Context-based	Arrow of Time		Benaim et al. (2020), Dwibedi et al. (2018), Destro and Gygli (2024), Donahue and Elhamifar (2024), Salehi et al. (2023), Wei et al. (2018), Wu and Wang (2021)
	Jigsaw		Kim et al. (2019), Lee et al. (2017), Liu et al. (2024b), Misra et al. (2016), Wang et al. (2022a), Xu et al. (2019a)
	Colorization		Ali et al. (2023), Dhiman et al. (2023), Jabri et al. (2020), Liu et al. (2024i), Vondrick et al. (2018), Wu et al. (2020b), Zhang et al. (2023e)
Contrastive	Negative Samples	MoCo (He et al., 2020)	Feichtenhofer et al. (2021), Han et al. (2020a), Kuang et al. (2021), Liu et al. (2021c), Ma et al. (2021), Pan et al. (2021b), Qian et al. (2021), Xu and Wang (2021), Yao et al. (2021)
		SimCLR (Chen et al., 2020c)	Badamdorj et al. (2022), Chen et al. (2021a), Han et al. (2020b), Jenni and Jin (2021), Sun et al. (2021a), Wang et al. (2020a), Yang et al. (2020a), Zhang et al. (2021b)
		CPC (Oord et al., 2018)	Akbari et al. (2021), Bagad et al. (2023) Dave et al. (2022), Li et al. (2021a), Miech et al. (2020b), Park et al. (2022a), Parthasarathy et al. (2023), Yang et al. (2021b)
		DIM (Hjelm et al., 2018)	Bai et al. (2022), Cai et al. (2022a), Feng et al. (2023), Gordon et al. (2020), Hjelm and Bachman (2020), Nan et al. (2021), Sameni et al. (2023), Sun et al. (2019a)
	Clustering	SwAV (Caron et al., 2020)	Coskun et al. (2022), Diba et al. (2021), Long and van Noord (2023), Toering et al. (2022), Wei et al. (2022b), Yan et al. (2020)
	Self-Distillation	BYOL (Grill et al., 2020)	Escontrela et al. (2023), Liu et al. (2022h), Morales et al. (2022), Recasens et al. (2021), Ranasinghe et al. (2022), Sarkar et al. (2023), Xiong et al. (2021), Zhang et al. (2022e)
		DINO (Caron et al., 2021)	Ding et al. (2024), Fan et al. (2023), Huang et al. (2024e), Huang et al. (2024b), Ponimatkina et al. (2023), Teeti et al. (2023), Wang et al. (2024c)
	Decorrelation	Barlow Twins (Zbontar et al., 2021)	Da Costa et al. (2022), Peh et al. (2024), Zhang et al. (2022e), Zhou et al. (2023b)
		VICReg (Bardes et al., 2021)	Bardes et al. (2023), Sun et al. (2023), Yang et al. (2023b), Yu et al. (2024c)

Table 3 continued

Learning	Task	Method	Video adaptations ¹
Masking	low-level targets	ViT (Dosovitskiy et al., 2020)	Girdhar et al. (2023a), Lin et al. (2022c), Piergiovanni et al. (2023)
		MAE (He et al., 2022b)	Feichtenhofer et al. (2022), Girdhar et al. (2023b), Huang et al. (2023a), Huang et al. (2023b), Ryali et al. (2023), Tong et al. (2022), Wang et al. (2023d), Wu et al. (2023c)
	high-level targets	BEiT (Bao et al., 2021)	Cheng et al. (2023), Fu et al. (2021), Li et al. (2023f), Tan et al. (2021a), Wang et al. (2022b)
	Teacher-based	data2vec (Baevski et al., 2022)	Li et al. (2023c), Lian et al. (2023)
		MaskFeat (Wei et al., 2022a)	Feichtenhofer et al. (2022), Mizrahi et al. (2023), Lin et al. (2023c), Pei et al. (2024) Stergiou et al. (2024), Wang et al. (2023f), Woo et al. (2023), Zhao et al. (2024c)

Signal-based works have also used the direction of motion flow over time (Runia et al., 2018) to count repetitions. Another set of methods approached VRC as a classification task over a finite set of maximum repetitions. Lu and Ferrier (2004) used dynamic parameters based on the Frobenius norm to classify changes corresponding to action end times. Zhang et al. (2021e) fused audio and video representations while Zhang et al. (2020a) used multiple cycles to refine the repetition count prediction. Li et al. (2024j) extracted action query features and classified the queries by their repetitions. In contrast to defining repetition counts as classes, Dwibedi et al. (2020) adopted a temporal self-similarity matrix (BenAbdelkader et al., 2004; Junejo et al., 2010; Körner & Denzler, 2013) to discover repetition periodicity. Subsequent methods have investigated embedding similarity matrices at multiple scales (Bacharidis & Argyros, 2023; Hu et al., 2022a), triplet contrastive losses (Destro & Gygli, 2024), and graph representations (Panagiotakis et al., 2018). Because embeddings of adjacent frames are highly similar, several recent methods have aimed to limit the discovery of correspondences in repetitions to poses (Ferreira et al., 2021; Yao et al., 2023), specific frames (Li & Xu, 2024; Zhao et al., 2024d), visual exemplars (Sinha et al., 2024), or language descriptions (Dwibedi et al., 2024). VRC remains a challenging task given the open-set nature and the lack of robust baselines in recent large-scale datasets (Dwibedi et al., 2024).

4.2.5 Future outlooks

Despite the great progress, using unified systems to generalize across tasks remains challenging. For example, STAD methods (Dai et al., 2021; Tirupattur et al., 2021) benchmarked on TAL perform lower than TAL-based models,

as their joint objective of localizing both *when* and *where* actions are performed is significantly more challenging to optimize. Similarly, despite the task similarities between TAL and VRC, standard TAL methods do not generalize to VRC as action interruptions and out-of-distribution categories cannot be effectively segmented (Hu et al., 2022a; Sinha et al., 2024). The recent introduction of unified VLMs for multiple video tasks, e.g., (Wang et al., 2024e) and their use as a feature extractor in subsequent works (Chen et al., 2024b), has shown a promising direction through the use of SSL. Training recipes typically include multiple stages of contrastive and masking pretext self-supervision objectives to allow the generalization of the model to multiple tasks. Training on SSL pretext tasks is a prominent scheme for many video-based models, as shown in Table 3. Context-based approaches rely on inherited spatiotemporal structural relationships in videos. Contrastive objectives are based on instance discrimination tasks, while masking tasks learn representation structures through completion. A possible direction of future research can be the unification of downstream objectives through relevant context, contrastive, and masking pretext tasks based on *the arrow of time*, relationships between task-specific embeddings, or clustering embeddings of semantically similar tasks.

4.3 Language semantics in videos

LLMs have achieved great success in Natural Language Processing (NLP) and have consequently been adapted for action understanding tasks. The relationships between learned context-rich semantic space and visual world attributes are useful for tasks such as caption generation (Seo et al., 2022; Sun et al., 2019b; Wang et al., 2024a), inferring scene

information (Anderson et al., 2018; Cheng et al., 2024), understanding the general context in highlight detection (Lei et al., 2021a), and instructional video learning (Miech et al., 2020b). Beyond their direct applicability to language-based tasks, they can incorporate vision encoders (Ashutosh et al., 2023a; Fu et al., 2021; Kahatapitiya et al., 2024; Song et al., 2024; Xu et al., 2021; Zellers et al., 2021) learning general and semantically-rich representations that can then be used as feature extractors in downstream tasks. However, notable challenges persist despite the popularity of vision-language semantic similarity pretraining for distilling context into video models.

4.3.1 Challenges

Visual and language information can provide partly complementary perspectives of a video. However, as information from each modality is often heterogeneous, specific representations may not be directly matched through cross-modal correspondence, *e.g.*, due to occluded objects or fine-grained visual details about the performance of the action. Such discrepancies can arise based on domain knowledge specificity or distribution patterns of the available data (Liang et al., 2024c). **Modality gap** (Liang et al., 2022d), shown in Figure 6, is a phenomenon that arises in VLM training in which embeddings of each modality are represented in distinct low-variance regions in the embedding space. In VLMs trained with cross-modal information maximization (Bain et al., 2021; Lei et al., 2021b; Li et al., 2020a, 2022a; Zhu & Yang, 2020) this effect becomes stronger with the enforcement of strong coordinate restrictions based on positive and negative cross-modal pairs. This is also relevant to difficulties in the **cross-modal context alignment** over local elements for tasks with available ground-truth pairs and global representations for tasks without vision-language pairs. In both cases, aligning language and vision context information at either the word/object level or over groups of instances in the embedding space provides a significant challenge in optimization. For generative tasks, this can also lead to difficulties in **modality-specific generation**. Generating semantic-rich data based on relationships from auxiliary modalities with ambiguous correspondence can impact conditional, stochastic, or auto-regressive generation.

4.3.2 Vision-language retrieval

Video retrieval sources relevant videos from a dataset based on an input query in natural language. As shown in Figure 7, research works can be categorized into instance- and semantic-based.

Image methods have explored cross-view ranking (Wang et al., 2016a), language to visual attention (Torabi et al., 2016), or visual features as embedding targets for language

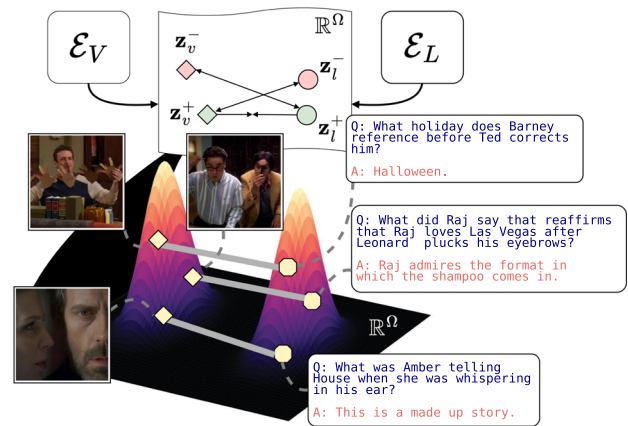


Fig. 6 VLM modality gap. Given video encoder $\mathcal{E}_V$ and text encoder $\mathcal{E}_L$, video and text are embedded to $\mathbf{z}_v^+$ and $\mathbf{z}_l^+$ in a joint embedding space $\mathbb{R}^\Omega$. VLM objectives align both $\mathbf{z}_v^+$ and $\mathbf{z}_l^+$. Contrastive approaches (Chen et al., 2020c; Oord et al., 2018; Xu et al., 2021) additionally maximize the distance between negative vision-language pairs: $(\mathbf{z}_v^+, \mathbf{z}_l^-)$ and $(\mathbf{z}_v^-, \mathbf{z}_l^+)$. Despite high-level semantic similarity, relevant modality-specific information that is not transferable across modalities can lead to a *modality gap* over embeddings. Videos sourced from Lei et al. (2018)

encodings (Dong et al., 2018). Early adaptation of visual-language approaches to videos have used image-text-video triplets (Otani et al., 2016), and related parts of speech to objects and actions (Gabeur et al., 2020; Xu et al., 2015c).

Instance-based approaches use a binary score function to rank correspondences. This formulation assumes only a single relevant caption/video for each video/caption. Refinements to this objective have been made through visual-language binding with the inclusion of parts-of-speech in target captions (Wray et al., 2019) and dual object-text and action-text models (Liu et al., 2019; Mithun et al., 2018). A number of methods have studied vision-language pretraining approaches (Ge et al., 2022b; Lin et al., 2022b; Xue et al., 2022). Ge et al. (2022b) related verbs and nouns to questions and video segments. Xue et al. (2022) studied the correspondences between keyframes and all video frames, subsequently contrasting the keyframe-fused video features to language embeddings.

Semantics-based. A more challenging task is to retrieve images based on shared semantics to query images (Gordo & Larlus, 2017). Semantic-based approaches primarily use triplet losses that contrastively regress between text queries and corresponding positive and negative visual inputs. These methods are based on the similarity between every (video, and caption) pair. Video retrieval works have studied this through either a contrastive objective based on a support set of videos with similar action categories (Patrick et al., 2020) or a semantic similarity scoring function for videos from the same category (Wray et al., 2021). Recently, Kim et al. (2024c) have utilized prior knowledge in retrieving text fea-



Fig. 7 Video retrieval tasks. (a) Instance-based retrieval returns only a *single video* corresponding to a search query. (b) Semantics-based retrieval returns a *ranking score* corresponding to each video's relevance

to the search query. (c) Temporal Sentence Grounding (TSG) receives *video segments from queries* and returns the start and end time per segment. Videos sourced from Xu et al. (2016)

tures based on embedding correspondences to similar visual features. Chun et al. (2021) proposed probabilistic representations of visual features to accommodate multi-query relevance. Similarly, Li et al. (2023d) created an object-phrase and event-phrase prototype-matching framework to enforce relations between high-level concepts across modalities. Hao and Zhang (2024) employed an uncertainty estimate based on the Wasserstein distance between source and target domains of text-vision pairs.

Temporal sentence grounding (TSG). TSG (Regneri et al., 2013) localizes moments in videos based on provided natural language queries. Compared to retrieving entire videos, TSG only retrieves relevant segments from a video based on queries. The task closely relates to TAL as also shown by the overlapping works (Gao et al., 2017a) jointly exploring the two tasks. However, in contrast to TAL, TSG requires both natural language reasoning between query and answer, and language-vision reasoning with query-video and answer-video relevance. Following Gao et al. (2017a), a broad formulation of TSG includes a visual-language semantic alignment between videos and sentences with a regression loss used for temporal sentence localization.

Early approaches have explored region proposals (Chen et al., 2018a; Qu et al., 2020; Liu et al., 2018a), ranking (Escorcia et al., 2019), distance-based joint vision-language embeddings (Hendricks et al., 2017; Rohrbach et al., 2016), and cross-modal graph representations (Liu et al., 2022a; Zhang et al., 2019a). As the association of visual and text features can be performed at multiple levels of abstraction, subsequent methods have learned correspondences over multiple proposals (Xu et al., 2019b), local and global information (Jiang et al., 2019a; Mun et al., 2020), and word/sentence-level cues (Hao et al., 2022). Zhang et al. (2021c) used language-guided highlighting by cross-attending text features to multi-resolution video features. Another important

aspect of discovering associations between the two modalities is their conditionality, as visual aspects should depend on the descriptions. Approaches have explored step-wise fusion of language key and value tokens (Cao et al., 2021), localizing relevant video features based on text embeddings (Yang et al., 2022a), matching video segments and text features contrastively (Flanagan et al., 2023), and enforcing similarity between sequential tokens (Qian et al., 2024). Ge et al. (2019) used both instance-based vision-language embeddings and general category representations to calculate an actionness score and location offset. Other approaches have fused context from global and local temporal resolutions (Liu et al., 2021a), grounded cues from anchor frames and boundary proposals (Wang et al., 2020b), related unimodal and cross-modal representations (Nan et al., 2021), and adopted instance-relevant positional information (Gu et al., 2024a). Goletto et al. (2024) explored TSG for language hand-object interaction queries.

4.3.3 Video Captioning

A long-standing challenge in computer vision is the generation of high-level descriptions in language. In contrast to retrieval tasks that depend on a fixed vocabulary, captioning is a generative task. Starting from matching a small corpus of words to objects in images (Barnard & Forsyth, 2001; Barnard et al., 2003), current works in the image domain are capable of generating diverse and detailed image descriptions (Mokady et al., 2021; Alayrac et al., 2022). Video captioning includes further challenges as the appearance of objects and the context of scenes change throughout the video. Given the temporal extent of videos, captioning tasks can be divided into two categories (Table 4).

Single video captioning. A large number of works have studied the direct extension of image captioning to video

Table 4 Video captioning papers grouped by target task and overall approach. Tasks are grouped by the generation of single or dense captions and the specialization to coherency with visual storytelling. Approach denotes architectural and model choices

Task	Approach	Works
Single video captioning	CNN+LSTM	Aafaq et al. (2019), Chen et al. (2017a), Gan et al. (2017), Pan et al. (2017), Wang et al. (2018a), Zheng et al. (2020b)
	Trnsf-based	Lin et al. (2022a), Shen et al. (2023b), Yan et al. (2023a)
	VLM	Chen et al. (2024e), Seo et al. (2022)
Dense video captioning	Region proposals	Deng et al. (2021), Iashin and Rahtu (2020a), Iashin and Rahtu (2020b), Krishna et al. (2017), Li et al. (2018d), Mun et al. (2019), Shi et al. (2019), Wang et al. (2018b), Zhou et al. (2018c)
	MIL	Chen and Jiang (2021), Shen et al. (2017)
Visual storytelling	VLM	Li et al. (2019b), Yu et al. (2021), Xiao et al. (2022), Han et al. (2023b), Han et al. (2023a), Han et al. (2024)

with single captions. Given a video clip and a corresponding caption, a general formulation of a single video captioning objective would be the minimization of the log-likelihood of the caption conditioned on the video.

Initial efforts (Guadarrama et al., 2013) used semantic hierarchies with word selection through decision tree nodes. Following methods (Aafaq et al., 2019; Chen et al., 2017a; Gan et al., 2017; Pan et al., 2017; Wang et al., 2018a) used encoder-decoder architectures that combined CNNs' visual feature extractors and recurrent architectures (RNNs or LSTMs) to generate textual descriptions. To focus on object semantics, Aafaq et al. (2019); Zheng et al. (2020b) included object detector embeddings. The improved context size of transformers has enabled more recent approaches to explore spatio-temporal dynamics in videos and cross-modal relationships. Transformer approaches have included language supervision over hierarchies (Ye et al., 2022) and token masking (Lin et al., 2022a; Shen et al., 2023b; Yan et al., 2023a). Seo et al. (2022) used a VLM with the video encoder and language decoder trained jointly on a reconstruction loss. Recently, Chen et al. (2024e) explored knowledge distillation from multiple VLM models to generate captions. Majumder et al. (2024) jointly learned a viewpoint ranking model for video captioning in multi-view settings.

Dense video captioning. Dense video captioning approaches generate multiple captions and temporally ground them to corresponding video segments. This is a significantly more challenging task as distinct video segments need to be localized to generate corresponding captions. Early works on event localization (Krishna et al., 2017; Li et al., 2018d; Shi et al., 2019; Wang et al., 2018b; Zhou et al., 2018c) were based on proposal modules from extracted video features. To learn vision-language correspondence explicitly for regions of interest, Zhou et al. (2018c) used proposals as masks for visual and language embeddings. Other approaches improved proposal generation by using their sequential occurrence as a prior (Mun et al., 2019), deployed fea-

ture clipping based on proposals (Iashin & Rahtu, 2020a, b), refined general captions for each proposal (Deng et al., 2021), and used unique CLIP properties to generate distinct captions (Perrett et al., 2024). Overall, proposal-based methods are optimized on a loss that relates the proposal interval to the ground truth segment and a captioning loss. As ground truth proposals require exhaustive annotation efforts, more recent works have focused on proposal-free approaches. Shen et al. (2017) used Multi-Instance Learning (MIL) in which word instances are assigned to bags. They used a binary objective to separate positive bags in which at least one instance corresponds to a target word and negative bags in which no instance contains the target word. MIL has been a building block in subsequent weakly-supervised approaches (Chen & Jiang, 2021). Further works (Yang et al., 2023a; Ren et al., 2024a) have also used sequence-to-sequence modeling with learnable time tokens for visual-language relations. Mavroudi et al. (2023) combined instruction learning to model the sequentially of video captioning. Islam et al. (2024) used a two-stage autoregressive approach that first generates dense captions for short clips and then cross-attends them to visual features over longer segments to generate longer captions. Zhou et al. (2024) aimed at efficiency improvements by compressing frame-instance visual features to clusters.

Visual storytelling. A recently introduced challenging task that is gaining interest is the generation of coherent sentences for sequential videos (Li et al., 2019b). To bridge cross-modal semantics, Yu et al. (2021) used a coherence loss for past, present, and future frames and contrastively pulled visual and language embeddings closer. A similar contrastive objective was used by (Xiao et al., 2022) alongside masking part of the visual input. Han et al. (2023b) trained a mapping module to project joint CLIP visual features, audio descriptions, and subtitles to an LLM input space to generate captions. Following efforts proposed additional refinements in the pipeline by injecting visual and caption embeddings over multiple LLM

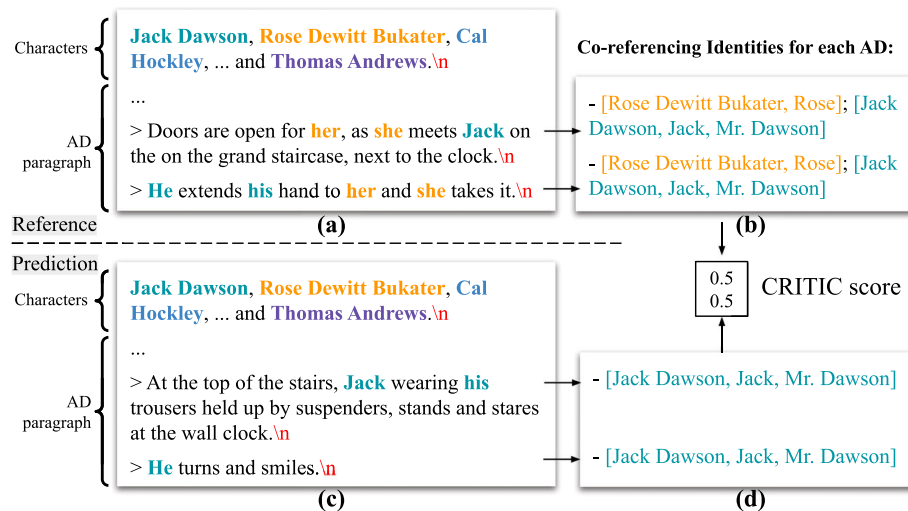


Fig. 8 CRITIC metric for visual storytelling. Identities are obtained from character lists and descriptions fed to a co-referencing model. CRITIC (Han et al., 2024) is calculated as the IoU between predicted and reference identities

layers (Han et al., 2023a), using exemplars (Han et al., 2024), and using character-based prompting (Xie et al., 2024). The CRITIC metric, shown in Figure 8, was recently introduced by Han et al. (2024) to measure the conceptual alignment of the generated sentences.

4.3.4 Video Question Answering (VideoQA)

A widely-used benchmark for VLM models is the utilization of visual context to answer natural language questions (Antol et al., 2015; Goyal et al., 2017b). In contrast to video captioning, it requires understanding parts of objects and the temporal extent of relevant answers. Depending on the task setting, answers can be obtained from multi-choice QA or as a global answer in open-end QA.

In a multi-choice QA setting, given a video and a question, the goal is to learn a mapping that returns an answer from a set of possible answers. In open-end QA settings, the answer is instead generated from a model conditioned on the video and question. VQA methods can be divided into two broad groups (Figure 9).

Scene-graphs. Early VideoQA approaches were based on either graph representations (Jiang & Han, 2020; Tu et al., 2014) or on each modality's heterogeneity. Huang et al. (2020a) used object and location-based graph embeddings to relate visual and text features with a cross-modal similarity matrix. Graph representations have also been created from hierarchies of objects and their interactions (Dang et al., 2021) as well as over multiple frames (Liu et al., 2021b). Alternative approaches defined scales from multiple graph convolution resolutions to relate cross-scale interactions (Guo et al., 2021) or from subgraphs to capture static and dynamic scene objects (Cherian et al., 2022). Park et al.

(2021a) created appearance, motion, and question graphs, learning conditionality by propagating nodes across graphs. Graph representations have also been learned contrastively (Xiao et al., 2023) from positive and negative pairs of video snippets and answers.

Multimodal memory. Another set of methods aims to memorize relations between visual and text features across time. Initial efforts integrated additional memory modules in LSTMs (Jang et al., 2017; Xu et al., 2017a; Zeng et al., 2017). Attention-based approaches (Ye et al., 2017) combined modality-specific memory modules (Fan et al., 2019) and memory-sharing modules to cross-attend motion and appearance (Gao et al., 2018; Li et al., 2019c). Several works (Gao et al., 2023; Li et al., 2023g; Yang et al., 2022b; Xue et al., 2023) have used a single model with concatenated language and vision tokens to predict answers to queries. Recent approaches have adapted large VLMs for VideoQA. Yu et al. (2023b) used a two-stage dual-VLM to first localize video segments based on the video and question and then used only the selected frames and question to generate the answer. Similarly, Min et al. (2024) used a list of generated VLM captions describing scenes in videos as input to an LLM. The question was then passed as a prompt to discover the most relevant answer. However, recent efforts (Xiao et al., 2024) have also revealed that VLM-based approaches may produce answers based on spurious language correlations and not the visual context.

4.3.5 Future outlooks

Advancements in VLMs have enabled the recognition of actions based on their correspondence to a large lexical corpus. Building upon this correspondence, retrieval, cap-

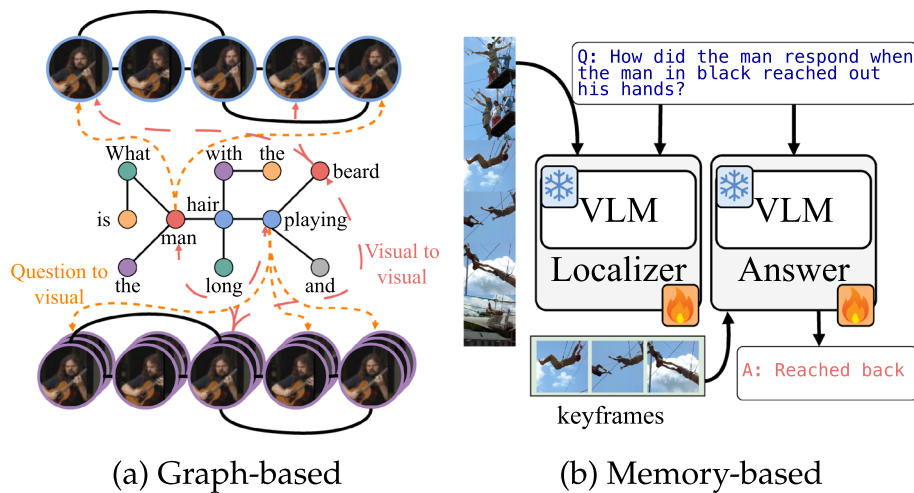


Fig. 9 VideoQA approaches. The graph-based approach in (a) is based on the method from Park et al. (2021a). The memory-based approach with a two-stage VLM in (b) is based on Yu et al. (2023b). Videos sourced from Xiao et al. (2021)

tioning, and question-answering models have moved beyond single-instance structural representations and toward the discovery of abstract cross-modal semantics. The increased model capacity provides opportunities for future lines of research.

Most VLMs strongly rely on linguistic associations that may not be relevant in vision instances (Rahmanzadehgervi et al., 2024). A possible alternative is to develop unified multimodal models that tokenize and encode video frames and images in the same manner with positional embeddings also encoding temporal relationships. Initial efforts by Jang et al. (2023) and Jin et al. (2024) have been promising. Another direction includes a better exploration of the objectives used. The majority of works train models on objectives (Chen et al., 2020c; He et al., 2020; Oord et al., 2018) or using downstream task adapters (Hu et al., 2021) which can enforce properties such as feature suppression (Chen et al., 2021c) and pretext granularity (Cole et al., 2022) despite aiming to maximize correspondence. Crafting better alignment objectives for cross-model representations and semantic relevance can benefit future VLM approaches.

4.4 Multimodal recognition

The recognition of actions or activities has been predominantly studied in the vision domain. In contrast, the auditory recognition of actions from sounds emitted by objects or actors and their interactions is more sparsely researched. This task presents distinct challenges as the sounds emitted by different objects or actions can be similar.

Time-frequency spectrograms have been a popular format for representing audio events in videos. Initial audio-based models have been built following image-based object recognition (Gong et al., 2021) or video classification (Kazakos et

al., 2021) CNNs. Attention-based audio methods have used convolutional features (Gulati et al., 2020; Kong et al., 2020) or image-pretrained encoders (Koutini et al., 2022) to attend over spectrogram patches. Approaches have also explored patch masking (Baade et al., 2022; Huang et al., 2022b), focused on salient sounds (Stergiou & Damen, 2023a), and adapted (Liu et al., 2022b) or compressed (Feng et al., 2024) spectrogram resolutions. More recently, the use of audio has gained attention in multimodal systems as it can provide supplementary information to both visual features and language context.

4.4.1 Challenges

The use of multiple modalities introduces several challenges. **Learning cross-modal dynamics** is a fundamental challenge of multimodal models as it aims to preserve heterogeneous properties of modalities while maintaining interconnectivity between modalities (Liang et al., 2022c). Fused embedding spaces (Girdhar et al., 2023a, 2022; Piergiovanni et al., 2023; Zhu et al., 2024b) effectively reduce modality-specific information and instead rely on learning a high level of abstraction, with lower heterogeneity and higher interconnectivity. In contrast, modality-specific embedding spaces (Gong et al., 2022b, 2023; Chen et al., 2024a; Recasens et al., 2021) are learned through cross-modal associations and rely on effectively transferring distribution across modalities, causing higher heterogeneity and lower interconnectivity. These paradigms are affected by **domain-specific noise topologies**. Based on a given task or data distribution, the discriminability of each modality differs. Noise naturally occurs based on environment settings, *e.g.* visual features are more relevant in daylight than in night videos. It can also be observed with instance-based occlusions or sensory

imperfections. Such topologies are important when developing reasoning structures over modalities (Gat et al., 2021).

Input representation reasoning can be defined as combining knowledge from the data and the structure of the objective. Compositional relationships between modalities can be established through concept hierarchies, temporal correspondence, or interactive states. Commonly, such structures are not available beforehand and are instead learned in an unsupervised manner.

4.4.2 Audio-visual models

As video and audio signals differ significantly, works have used two-step models to infer predictions. Two-step approaches extract video and audio embeddings first and then fuse either modality-specific predictions (Fayek & Kumar, 2020), embeddings from multiple modalities (Xiao et al., 2020), or they jointly attend vision and audio features for the final prediction (Gong et al., 2022b). More recently, architectures have tokenized and attended audio and vision jointly with multimodal learnable tokens (Nagrani et al., 2021), cross-modal attention (Jaegle et al., 2021), and modality gating (Xue & Marculescu, 2023). To account for models trained on unimodal tasks, Lin et al. (2023d) proposed cross-modal adapters to combine unimodal embeddings in multimodal tasks. Exploring the relevant audio and visual features with self-supervision has also been a learning paradigm of significant interest. Common embedding spaces can be useful for discovering correspondences in both unimodal and cross-modal retrieval (Arandjelovic & Zisserman, 2018; Wu & Yang, 2021), multimodal clustering (Hu et al., 2019), and sound source separation (Hu et al., 2022b; Mo & Morgado, 2023; Zhao et al., 2018). Token reconstruction through masking has also been used as a self-supervised pretraining task with a variety of training schemes, including concatenating masked tokens (Gong et al., 2023), multi-view masking per modality (Huang et al., 2023b), fusing a mixture of per-modality masked tokens (Guo et al., 2024b), combining modality-specific masked and unmasked embeddings (Georgescu et al., 2023), and using multiple masking ratios with siamese networks (Lin & Bertasius, 2024).

Variations in the relevance of visual or auditory signals depend on instances. A promising direction for integrating this into optimization is gradient blending (Wang et al., 2020c) which recalibrates per-modality losses. Other works explored multi-audio to single-visual scene correspondence with contrastive learning. This was done by utilizing joint semantic similarity in both modalities (Morgado et al., 2021), using active sampling to diversify negative sample selection (Ma et al., 2021), and by counterfactual audio and video pairs to enforce a relationship between multi-audio to single visual scenes (Singh et al., 2024). Enforced audio and vision steams

similarities can also be used to train models on incremental tasks (Pian et al., 2023).

4.4.3 Gaze and vision models

Gaze can be used as a saliency cue to direct attention or processing priority towards target focal areas (Itti et al., 2002). Several egocentric activity datasets (Huang et al., 2024c; Li et al., 2018c; Pan et al., 2023) record gaze as an additional modality. Based on gaze inputs, gaze fixations have been used to recognize objects being manipulated (Land & Hayhoe, 2001), and discover ways of interacting with them (Damen et al., 2016). Observing gaze patterns, including fixations and saccades, in conjunction with scene appearance has also been used to predict future gaze behavior (Huang et al., 2018b).

Because gaze and action are tightly coupled (Vickers, 2009), subsequent research has focused on action recognition with the additional availability of gaze information. For example, Fathi et al. (2012) probabilistically modeled the joint distribution of gaze target, scene objects, and action label. Min and Corso (2021) modeled gaze fixations as latent variables for activity classification. Xu et al. (2015b) used gaze to identify relevant actions in long videos, to summarize egocentric videos. More recently, gaze patterns have been used to identify deviations from expected executions of procedural activities (Mazzamuto et al., 2025).

Gaze has also been used to train image and video models with gaze-free inference. Liu et al. (2021e) learned to attend discriminative local features for zero-shot object identification. In general, the additional availability of gaze has shown improvements in training across egocentric video understanding tasks (Kapidis et al., 2023).

Several works have also addressed the opposite process of estimating salient regions likely to be looked at given an image or video. While initial works introduced bottom-up algorithms (Borji & Itti, 2012), later research increasingly considered higher-level information in predicting gaze targets (Judd et al., 2009; Torralba et al., 2006). In egocentric videos, it was shown that object presence and, in particular, object manipulation strongly directed eye gaze (Tavakoli et al., 2019). This notion has led to fusion approaches that first process either global scene context and local saliency independently (Lai et al., 2024a), or static and dynamic information as separate branches of a two-stream model (Lu et al., 2019).

Other methods focused on action- or task-dependent gaze estimation (Huang et al., 2018b). Li et al. (2018c) jointly determined gaze targets and the person's actions. Huang et al. (2020b) jointly modeled gaze-conditioned action recognition and action-conditioned gaze estimation in a single network. Chong et al. (2020a) recognized eye contact in egocentric videos. Models that predict gaze over subsequent frames have also been introduced (Zhang et al., 2017). Recently, auditory

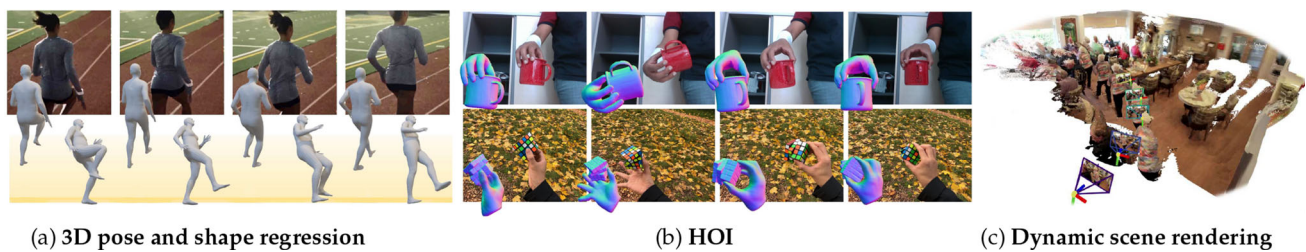


Fig. 10 4D video understanding tasks. (a) 3D human pose and shape regression takes as input monocular videos and produces expressive 3D representations. (b) Human/hand-object interactions predict aspects of human-object interactions, such as the contact area or the grasp. (c)

Dynamic scene rendering estimates the per-timestep geometry of scenes from sets of images. Figures sourced from Dwivedi et al. (2024); Fan et al. (2024); Zhang et al. (2025)

information was included to improve gaze predictions (Lai et al., 2024b).

Estimation of eye gaze in third-person perspectives additionally considers a person's body and head orientation (Chong et al., 2020b; Marín-Jiménez et al., 2021) to estimate the focus of their gaze. Recasens et al. (2017) extend the gaze prediction to targets that appear in subsequent frames by learning correspondences between views. Increasingly, gaze predictions in third-person videos consider multiple persons to include social context (Tafasca et al., 2024).

4.4.4 4D-vision methods

While the majority of the research has addressed 2D video and time as the third dimension, increasingly scenes are modeled in 3D, leading to the space-time-depth (4D) reconstruction of humans, objects, and scenes. We provide a visualization of such tasks in Figure 10 and discuss the main approaches and objectives per task below.

Pose and shape regression. Initial efforts for human pose and shape regression from monocular images primarily captured the overall body shape and pose (Allen et al., 2003, 2006; Loper et al., 2015). Additionally, finer details were modeled with either MANO (Romero et al., 2017) for hands or the Frank model (Joo et al., 2018) for faces. Unified parametric frameworks were later introduced to produce pose and shape details through keypoints (Hassan et al., 2019; Kolotouros et al., 2019; Pavlakos et al., 2019; Xu et al., 2020) or silhouettes (Kanazawa et al., 2018; Omran et al., 2018). Attention-based methods have also been used with either pseudo-ground-truths to align body-to-image projections (Joo et al., 2021; Li et al., 2022f; Moon et al., 2022) or with probabilistic representations (Li et al., 2024d; Sengupta et al., 2023; Stathopoulos et al., 2024; Zhang et al., 2023d). Recent methods have used quantized human poses as a prior (Dwivedi et al., 2024; Fiche et al., 2024).

3D human and object interaction (HOI). HOI tasks seek to capture human and object relations in 3D. They include estimations of dense human-object contact (Huang et al., 2024d;

Jiang et al., 2023; Nam et al., 2024; Yang et al., 2024b; Xie et al., 2022) through 2D image-based semantics and geometric correlations. Other approaches learned object affordances (Mo et al., 2021; Zhai et al., 2024) by mapping them to their shapes, human-object interactions, and spatial relations (Liu et al., 2023b; Xu et al., 2023c, 2025). Extensions to these tasks have also included human-human and human-scene interactions (Fieraru et al., 2020; Huang et al., 2024a; Yin et al., 2023b), self-contact (Fieraru et al., 2021; Muller et al., 2021), and predicting human and scene layouts (Huang et al., 2022a; Zhang et al., 2020b).

In hand-object interactions, methods used pre-defined object templates to estimate poses from 3D control points (Hampali et al., 2020; Tekin et al., 2019) and temporally-consistent sparse labels (Hasson et al., 2020; Liu et al., 2021d). Other template-based methods have explored object-centric grasp prediction from frames (Corona et al., 2020; Hasson et al., 2019), contact prediction from hand and object meshes (Grady et al., 2021; Zhu & Damen, 2023). Zhang et al. (2024a) used generative models to learn possible HOI sequences conditioned on sets of motion objectives and hand-object states. Template-free approaches have also gained interest for reconstructing novel objects in in-the-wild settings. The majority of these methods employ dual-branch models (Leng et al., 2023; Tse et al., 2022; Xu et al., 2023a) that directly predict pose, shape, and camera from frames (Dong et al., 2024; Pavlakos et al., 2024). Recently, Fan et al. (2024) jointly reconstructed hands and objects in monocular videos by refining initial structure-from-motion estimates through a per-frame-texture and shape regression objective followed by hand/object pose constraint fine-tuning.

Dynamic scene rendering. Rendering tasks estimate 3D scene structures from sets of 2D images for static scenes, and frames for dynamic scenes. Although well-established approaches such as structure-from-motion (Schonberger & Frahm, 2016; Teed & Deng, 2021), LSD-SLAM (Engel et al., 2014), and ORB-SLAM (Mur-Artal & Tardós, 2017) have been promising for static scenes, dynamic scene rendering remains an active challenge. Recent self-supervised

methods have jointly estimated depth, camera pose, and residual motion, with motion segmentation (Gordon et al., 2019; Godard et al., 2019; Kopf et al., 2021; Zhang et al., 2022f). Another set of methods focuses on 4D dynamic scene reconstruction by space-time optimization of 3D Gaussians (Chu et al., 2024; Lei et al., 2024; Liu et al., 2024c) to synthesize novel views over both space and time. As the joint learning of geometry and motion can be difficult to learn end-to-end, Wang et al. (2024b) introduced a point-map scene geometry approach in which, given a pair of images, per-image pixels are mapped to discrete 3D locations and then accumulated to a global point cloud. Model pre-training was conducted over cross-view tasks (Weinzaepfel et al., 2023). This approach has prompted further extensions through assigning pointmaps to single points in time (Zhang et al., 2025), coupling intermediate predictions (Li et al., 2024k), and including a depth estimation model (Lu et al., 2025). Other approaches have aimed to improve speed (Liang et al., 2024a), and jointly reconstruct scenes and recover human meshes (Liu et al., 2025).

4.4.5 Multimodal models

Video is a natural source of multimodal data. Apart from visual information, audio or textual descriptions can be used in tandem to provide additional signals for actions and events at different granularities. Multimodal learning has shown improvements in the generalizability of unsupervised models (Ngiam et al., 2011) over varying tasks (Paredes et al., 2012). An initial effort by Kaiser et al. (2017) aimed to create unified multimodal representations with modality-specific encoders and modality-binding decoders. A similar mixture-of-experts approach was also presented by Munro and Damen (2020) for unsupervised domain adaptation with a dual cross-domain source-target loss over modality pairs. Dai et al. (2022b) used sparse activations to train portions of a unified model on specific modalities and tasks. Multimodal transformers have introduced joint encoder paradigms. Akbari et al. (2021) used modality-specific heads to project outputs from a joint audio-text-video encoder trained with a contrastive loss over modality pairs from Miech et al. (2020b). Mixtures of modality-specific encoders and multimodal head/decoders have also been trained with masked tokens (Zellers et al., 2022), cross-modal attention blocks in the encoder (Recasens et al., 2023), ensembles of unimodal teachers (Radevski et al., 2023), and audio-vision projectors on top of LLM heads (Zhang et al., 2023a). Zhang et al. (2024d) fused features from different modalities to a Multimodal head at different training steps capturing cross-modal associations iteratively during training. Srivastava and Sharma (2024a) included meta tokens to represent modality dimensions and channels to embed modality-specific features in a common space. This was



Fig. 11 Early Action Prediction (EAP). Only the observable part of a video $\tau_{1,\rho}$ is used to predict the current action. EAP is challenging as the immediate future is often unpredictable especially when distinguishing between fine-grained actions, e.g., *taking courgette* or *taking carrot*. Video from Damen et al. (2022)

further adjusted (Srivastava & Sharma, 2024b) to also cross-attend joint-embedded features and unimodal features.

4.4.6 Future outlooks

Most current multimodal models rely on the availability of all modalities at the start. New models are re-trained when additional modalities are added. A promising direction would be to design adaptive models that integrate unseen modalities more efficiently (Ma et al., 2022b). This has been explored by recent methods by transferring seen to unseen modality distributions (Wang et al., 2023j), cross-attending over unseen modalities (Recasens et al., 2023), aligning unimodal and multimodal features in training (Zhang et al., 2023f), using modality-specific adapters (Lin et al., 2023d), and predicting missing modality features with learnable tokens (Kim & Kim, 2024). Learning adaptable models that can process inputs in new modalities at inference time not only benefits performance for specific tasks but also enables advancement in more general tasks such as online learning (Bottou, 1998), incremental learning (Schlimmer & Fisher, 1986; Utgoff, 1989), and federated learning (Konečný et al., 2016).

5 Predictions in ongoing actions

A critical aspect of video models regardless of the downstream task has been their ability to capture temporal information (Huang et al., 2018a). Learning temporal patterns can enable models to predict what is happening in a video, without seeing the full action or activity. We start by defining methods that provide semantic predictions on the action categories from partial observations in Section 5.1. We then discuss approaches that generate unobserved frames in Section 5.2. States of actions or objects can change at different times during the execution of an action. We explore groups of tasks that predict the states of objects and actions in Section 5.3.

5.1 Early action prediction

Early Action Prediction (EAP) assumes predictions made based on the observable part of an *ongoing* action being performed $\tau_{1,\rho}$ as shown in Figure 11. Several different lines of research have been explored to address relationships between partial action observation and high-level semantics.

5.1.1 Challenges

One of the main challenges that arise from partially observed videos is the **procedural proximity** in the execution of actions. As shown in Figure 11, there can be an overlap in the steps taken to perform similar activities. For the example shown, the difference between *take courgette* and *take carrot* is subtle, without significant motion variations. Instead, the distinction only becomes visually apparent at the end of the action. This challenge relates to the more general problem of **abstracted views** in which deterministic information relevant to the task is not always available. Although such fine-grained predictions may not be available, it is still possible to identify general categories for the actions being performed, e.g., *take <object>*. An **adjustability requirement** is thus introduced as part of EAP to address the intrinsic uncertainty in partial observation.

5.1.2 Approaches

Three main groups of approaches can be identified for EAP. We visualize these groups in Figure 12.

Probabilistic modeling. A large portion of the EAP literature has originally been based on probabilistic modeling of action classification from partial observations (Cao et al., 2013; Hoai & De la Torre, 2014; Li et al., 2012; Li & Fu, 2014; Ryoo, 2011). Ryoo (2011) used a bag of words based on feature distributions. This division into segments has been relevant in subsequent approaches that used sparse coding (Cao et al., 2013), max-margin (Hoai & De la Torre, 2014), and scoring functions (Li et al., 2012; Li & Fu, 2014) to infer the action likelihood. More recent probabilistic approaches include the use of hyperbolic representations (Surís et al., 2021) for hierarchical predictions of actions. The ambiguity of future predictions has also been explored with the generation and subsequent selection of multiple future representations (Chen et al., 2022b).

Temporal ordering. A different line of works explored EAP based on the temporal evolution of the action. The arrow of time (Pickup et al., 2014) can provide a strong signal to associate the procedural understanding of actions with high-level categorical semantics (Misra et al., 2016; Zhou & Berg, 2015). Xu et al. (2015d) formulated EAP with an auto-completion objective, matching candidate futures to a partial action observation query. The predictability of par-

tial observations can be difficult in instances where there are visual similarities in the performance of actions. To address this, approaches have either used multiple temporal scales (Kong et al., 2014), created key-value memories of representations (Kong et al., 2018), or propagated the features' residuals over time (Zhao & Wildes, 2019). More recent approaches have used temporal graph representations (Wu et al., 2021c,d), contrastive learning over partial observations of the same action (Wang et al., 2023g), or aggregated attention over temporal scales (Stergiou & Damen, 2023b), and relevant space-time regions (Rangrej et al., 2023).

Knowledge distillation. Transferring class knowledge (Park et al., 2019) from models trained on the full videos can be an effective technique for refining predictions from partial observations. Cai et al. (2019), Fernando and Herath (2021), and Wang et al. (2019a) used learned representations of the full observations as targets to optimize for partial observations. Further methods (Hou et al., 2020) have refined this approach with the inclusion of motion sequentiality to learn soft targets and regress model predictions. In a similar effort, (Xu et al., 2019c) and (Zheng et al., 2023) integrated an adversarial objective for generating representations for the non-observable parts. Similarly, Xu et al. (2023d) learned to reconstruct representations of full observations with a masked autoencoder (He et al., 2022b). Other works have fine-tuned expert heads for each action category (Foo et al., 2022) or learned by focusing on videos with distinct visual features (Hu et al., 2018).

5.1.3 Future outlooks

Although EAP remains a challenging task to be explored further, some future directions can be envisioned. First, current EAP evaluation protocols are based on fixed-length temporal occlusions of parts of videos. However, this offline evaluation varies significantly from the intended **real-time use** of these systems in which singular models are deployed in video streams. EAP methods should instead be built and evaluated in real-time settings in which factors such as latency and inference speeds are crucial.

A second direction of future research is the exploration of **multi-person action prediction** for group activities. This is a significantly more complex task as it not only requires predicting the intentions of individuals but also general group goals. Such approaches will also have direct application to more general fields such as robotics, security, and augmented reality.

5.2 Frame-level prediction

Related to EAP, Video Frame Prediction (VFP) aims to reconstruct future frames of ongoing actions from partial observations. Although high-level semantics such as the level

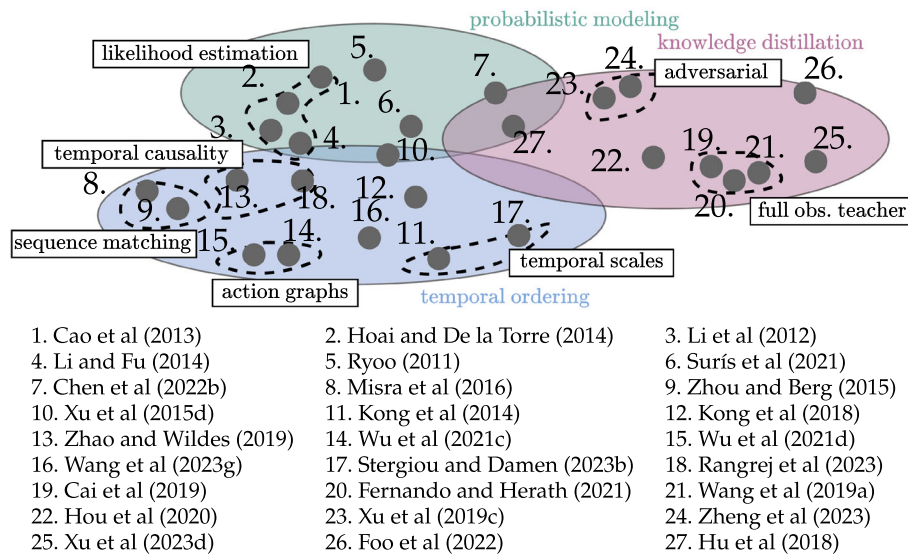


Fig. 12 EAP methods grouped by approach. The three main clusters are colored. Smaller subgroups are denoted with dashed lines. The positioning of the works represents an abstract proximity of the research idea to other seminal works

of semantic abstraction to describe the observed action are not learned, VFP still requires relating the consequentiality of motions and intended action to the reconstruction of subsequent frames.

5.2.1 Challenges

The **metric-based evaluation** of VFP approaches is done deterministically as VFP aims to predict raw pixel values of future frames. Metrics such as Peak-Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) have been widely adopted to quantify VFP performance. However, they do not evaluate the correctness of high-level scene dynamics or the consistency of the scene. A number of image-based statistics adjusted to video (Czolbe et al., 2020; Ding et al., 2020; Zhang et al., 2018) and video-specific statistics (Hou et al., 2022; Li et al., 2019a) have targeted such shortcomings by including comparisons in embedding representations between predicted and ground truth frames. The usability of these metrics still leaves room for exploring the robustness of quality validation approaches further.

Similar to EAP, **future scene dynamics predictions are stochastic**, with varying levels of complexity. Most VFP objectives are based on the changes in the pixel distributions between frames without explicit definitions of optimization criteria for a comprehensive understanding of physics or object structures within scenes. This hinders the prediction capabilities over longer temporal windows and complex scenes with fast motions (Ming et al., 2024).

5.2.2 VFP methods

We identify three groups of VFP approaches.

Sequential adversarial predictions. A significant portion of VFP works has been based on sequential frame generation (Castrejon et al., 2019; Chaabane et al., 2020; Chang et al., 2021, 2022; Chen et al., 2017b; Guen & Thome, 2020; Hwang et al., 2019; Jin et al., 2020; Liang et al., 2017; Villegas et al., 2018; Wang et al., 2018f; Wu et al., 2021b). These approaches use recursion to generate representations or predictions in an autoregressive manner. One line of methods (Chen et al., 2017b; Jin et al., 2017) focused on the correspondence of objects between frames to guide the generation of the next frames. Castrejon et al. (2019) used similar adversarial guidance by fusing context information from previous frames. Additional supervisory signals included motion flow (Liang et al., 2017), partial differential equations (Guen & Thome, 2020), and embeddings over multiple temporal resolutions (Gao et al., 2022b). Another line of approach (Chang et al., 2021; Villegas et al., 2018; Wang et al., 2018f) has included long-term memory connections to discover causalities from frames over longer temporal windows. Park et al. (2021b) incorporated time dynamics for VFP with the inclusion of Ordinary Differentiable Equations (ODE). Davtyan et al. (2023) used ODE with the previous frame as the initial condition and integrated the vector field from Flow Matching (Lipman et al., 2023) to predict the next frame.

Parallel multi-frame synthesis. In contrast to the sequential reconstruction of future frames, approaches have also generated multiple future frames in a single step. One of the first efforts for multi-frame prediction (Liu et al., 2017b) used a multi-frame per-pixel optical flow vector with fur-

ther adaptations including multiple scales (Hu et al., 2023). Attention-based architectures have also been used for parallelization of frame prediction by introducing encodings of context for frame prediction attended over temporal patches (Tan et al., 2023a; Ye & Bilodeau, 2023), conditioning the generation based on short-term representation variations (Hu et al., 2023; Smith et al., 2024), using multiple motion and appearance scales (Zhong et al., 2023a), and reducing inference speeds (Ye & Bilodeau, 2022; Tang et al., 2024). As an extension to spatiotemporal attention, Nie et al. (2024) used a triplet module to attend across all dimensions of the video sequentially.

Probabilistic generation. A final group of approaches studied the reconstruction of future frames probabilistically. Babaeizadeh et al. (2018) and Denton and Fergus (2018) learned a probabilistic variational model on the stochasticity of the video to generate frame predictions. Wang et al. (2020d) models the perceptual uncertainty in future frames with a Bayesian framework with different weights assigned to future prediction candidates. Diffusion-based models (Dhariwal & Nichol, 2021; Ho et al., 2020; Rombach et al., 2022) have been applied to a multitude of generative approaches for VFP (Gu et al., 2024b; Höppe et al., 2024; Shrivastava & Shrivastava, 2024; Voleti et al., 2022; Ye & Bilodeau, 2024; Zhang et al., 2024f). These methods gradually transform a complex distribution into unstructured noise and learn to progressively recover the original distribution from noise.

5.2.3 Future outlooks

The scarcity of **high-resolution datasets** remains a limiting factor for the performance of VFP models. More varying data distributions in terms of motions in scenes, sharpness, and blur can enable current approaches to learn across diverse video conditions. Recently, efforts (Stergiou, 2024; Xue et al., 2022) have aimed to provide high-resolution videos for several tasks. Future VFP approaches can benefit from training on these datasets.

A point of improvement for future works is the **inclusion of world knowledge** to enable predictions based on abstractions of the scene dynamics. The recently introduced term *Stochastic Inverse Problems* defines a broad family of problems relating to predictions from partial observations (Spielberg et al., 2023; Tewari et al., 2023). These approaches aim to permeate knowledge of physical underlying processes throughout training.

5.3 State changes

Another set of action prediction tasks includes modeling the state changes in the environment, actions, objects, or execution speeds. Tracking, inferring, and reasoning in these tasks

come with new sets of changes. An overview of the tasks' objectives is visualized in Figure 13.

5.3.1 Challenges

Understanding the sequentiality in videos is a central part of human perception. The translation of this to computer vision tasks remains challenging (de Boer et al., 2023). A central challenge relates to the **object state variability** as actions affect the visual appearance of objects in terms of shape, visibility, or perspective. Another set of challenges concerns actions with **non-rigid temporal boundaries**. Actions may not be easily distinguished from backgrounds or their execution may overlap with other actions. This presents **ambiguities in the action progress** as the completion of an action, or part of it, might also involve another action. Finally, the typically weak relation between visual input and the high-level semantic interpretation thereof complicates the training of robust, general models.

5.3.2 State-based tasks

We discuss six main tasks based on object and action state changes in their objectives. These include tracking the progress of actions (**action progress prediction**), defining start-end times (**action progress prediction**), temporally aligning action phases (**video alignment**), inferring action goals (**visual abductive reasoning**), tracking object appearance changes (**object state change detection**), and localizing relevant objects (**active object detection**).

Action Progress Prediction (APP). Actions can be understood by procedural sets of motions performed towards an intended goal as shown in Figure 13a Vaina and Jaulent (1991) suggested that understanding the state and progress of the action at different times can provide a holistic understanding of the intent and objective. In machine vision, an initial APP approach (Fathi & Rehg, 2013) used local descriptions to model per-frame state changes. (Kataoka et al., 2016) used a descriptor to discover transitional actions within activity sequences. Xiong et al. (2017) introduced a score function to distinguish between actions based on learned distinctive parts. Becattini et al. (2020) used actor and scene context information as an additional supervisory signal for APP. Price et al. (2022) expressed the progress of multiple actions through threads of activities that can overlap, a common situation in long procedural videos. Shen and Elhamifar (2024) causally attended videos to define a task graph for APP over each action. More recently, generative approaches (Souček et al., 2024b) using conditional control (Zhang et al., 2023c) and procedural knowledge (Ashutosh et al., 2023b; Zhou et al., 2023a) have been used to generate keyframes of changes.

Another line of research works (Heidarivinchek et al., 2016, 2018) is aimed at localizing the moments that actions



Fig. 13 Tasks relating to object and action state change. Each of the presented tasks can involve additional objectives

are completed. The speed of action completion or state changes in actions has also been studied in the context of skill determination (Doughty et al., 2018) or their semantic correspondence to textual adverbs (Doughty et al., 2020; Doughty & Snoek, 2022; Moltisanti et al., 2023). Scoring approaches (Tang et al., 2020b) have been used to study the procedural execution of actions in the context of quality assessment. Adjacent tasks such as video captioning and

action classification have also been integrated into multi-task settings (Parmar & Morris, 2019).

Event Boundary Detection (EBD). Different from the related well-studied task of action localization, EBD (Shou et al., 2021) localizes event changes in videos *regardless of the action classes*, shown in Figure 13b. Aakur and Sarkar (2019) proposed a self-supervised objective in which their model is initially trained to reconstruct subsequently observed features. (Shou et al., 2021) used a self-similarity metric to

determine event boundaries by relating encoded frame features. Further, EBD approaches (Mounir et al., 2023) have studied hierarchies of video events. Recently, Eyzaguirre et al. (2024) explored the detection of event starts from natural language queries.

Video Alignment (VA). As the performance of individual parts of actions can vary, video alignment, shown in Figure 13d, aims to temporally match key moments in the execution of the same action across videos. Initial efforts, motivated by temporal coherence (Goroshin et al., 2015; Fernando et al., 2017; Zhang et al., 2023b), have studied VA based on Canonical Correlation Analysis (CCA) (Andrew et al., 2013) or by contrastively creating joint representations from multiple viewpoints (Sermanet et al., 2018). Dynamic Time Warping (Sakoe & Chiba, 1978) is an algorithm that aligns variable length signals, and it has been adopted for VA (Chang et al., 2019; Dvornik et al., 2021; Hadji et al., 2021). A more recent self-supervision objective (Dwibedi et al., 2018) for VA is to train a video model to project per-frame embeddings in pairs of target videos by matching embeddings of one video to the nearest neighbor embeddings of the other. This approach was extended in subsequent works with the inclusion of context from the entire video (Haresh et al., 2021), anchor frames to align redundant frames (Liu et al., 2022d), embeddings from text (Epstein et al., 2021), and regularizers based on the correspondence to repetitions of the same action (Donahue & Elhamifar, 2024).

Visual Abductive Reasoning (VAR). A key element in action understanding is recovering the intended goal. High-level reasoning of events has initially been considered in hierarchies with rule-based approaches (Hakeem & Shah, 2004) or the conditionality of action occurrences across different levels (Albanese et al., 2010). Pei et al. (2011) detected atomic actions with graph representations to decompose complex events. VAR (Liang et al., 2022a), shown in Figure 13c, is the vision-language task that uses characteristics of partial observations as a premise and requires formulating an explanation. Other VAR works have modeled intention by contrastively learning visual and language context (Li et al., 2023b), modeling timelines for news story understanding (Liu et al., 2023a), and forecasting actions by multimodal inputs (Zhu et al., 2023). Evaluation of VAR models has also been studied in counterfactual vision-language pairs (Park et al., 2022b) similar to text-only tasks (Ippolito et al., 2019; Huang et al., 2020c).

Object State Change Detection (OSCD). Many actions alter the appearance or state of objects. OSCD approaches associate visual changes to changes in the states of objects in the scene, as shown in Figure 13e. Efforts (Alayrac et al., 2017; Damen et al., 2014; Liu et al., 2017a; Zhuo et al., 2019) have initially focused on state modifications that do not involve significant appearance changes, *e.g.*, open/close door or fill/empty cup. Hong et al. (2021) proposed

a reasoning-based approach defining a triplet of complexities for single- and multi-step transformations with additional viewpoint changes. Other reasoning-based approaches include the use of language (Xue et al., 2024) and visual exemplars of start and end states (Souček et al., 2022). OSCD has also been studied in combination with other tasks including cross-state object segmentation (Yu et al., 2023a), cross-action relevance (Souček et al., 2024a), or inspired by state-disentanglement for images (Gouidis et al., 2023; Nagarajan & Grauman, 2018; Saini et al., 2022), generating start and end states by given context and scene prompts (Souček et al., 2024b; Saini et al., 2023).

Active Object Detection (AOD). Actions can include multiple objects during their execution. Overviewed in Figure 13f, AOD localizes the objects relevant to the currently performed atomic action with bounding boxes. This task has recently gained interest as scenes can often be cluttered (Ragusa et al., 2021) or a varying number of objects can be used for a single action (Miech et al., 2019). Nagarajan et al. (2019) specifically focused on localizing the human-object interaction areas defining focal points of importance during the execution of actions. (Fu et al., 2022) introduced a voting module over potential bounding boxes corresponding to the active object. Kim et al. (2021a) used a parallelized model to detect instances and subsequently hand-object interactions. Yang and Liu (2024) used scene context from text to define plausible interactions with target objects for AOD.

5.3.3 Future outlooks

Similar to recent works in robotics that create goal-based policies (Kun et al., 2024; Wang et al., 2023c), state-understanding vision models require a holistic understanding of actions given a limited **availability of arbitrary object states**. Conceptually, the execution and changes in objects can be similar for different actions, *e.g.*, mixing a cake mix and whisking eggs. Enforcing better learning objectives in order to improve this semantic correspondence can create more generalizable models with a better understanding of the physical world. In addition, the inclusion of cues from **supplementary modalities**, such as audio, can improve tasks that require relating parts of videos, as correspondences should be discoverable beyond the visual domain. This also presents the potential for creating general-purpose models based on **abductive reasoning** pretext objectives that can then be applied to various downstream tasks.

5.4 Anomaly detection

Video Anomaly Detection (VAD) is the task of detecting unexpected actions or events in videos that deviate from predictable behaviors. Anomalies are detected through either

explicitly classifying a pre-set number of actions in **close-set** settings or learning robust representations of the expected actions in **open-set** settings.

5.4.1 Challenges

VAD depends on strict binary annotations of normal and abnormal sequences, with most evaluation benchmarks including **limited definitions**. Even in the open-set settings, robust definitions are required for the target normal sequences. This prevents the creation of models that can estimate correctly unseen normal sequences based on their visual proximity to other actions. Although recent-continual-learning approaches have also been proposed for VAD (Bugarin et al., 2024), their applicability remains sparse. Another significant challenge for VAD models relates to their applicability. With their intended use in continuously operating surveillance systems, current approaches only **partially use temporal context** to infer predictions. Only a small number of approaches currently study the long-term effects of actions. Most datasets also only include modest temporal resolutions, limiting the exploration of context over longer time segments.

5.4.2 Detecting anomalies in videos

We identify two main approaches for the detection of anomalies in videos.

Close-set. Anomalies can be discovered by *close-set* tasks that aim to model both normal and abnormal sequences. Sultan et al. (2018) used Multiple Instance Ranking (Dietterich et al., 1997) to define positive groups that include videos with at least a single abnormal segment and negative groups of normal videos. The objective is to maximize the score between the assigned positive and negative groups. Subsequent efforts have built upon MIL with learned features (Dubey et al., 2019), or pseudo labels (Feng et al., 2021a). Approaches have also aimed to improve upon MIL's reliance on the dominant negative instances. Zhang et al. (2019c) integrated inner-group sampling, Pu et al. (2024); Zhu and Newsam (2019) used temporal weighting, and Tian et al. (2021a) maximized the separability between normal and anomalous representations. With a similar goal, the use of multiple temporal pretext tasks (AlMarri et al., 2024; Georgescu et al., 2021) and temporal scales (Li et al., 2022b) have also been explored. Chen et al. (2023b) used a contrastive objective between representations of normal and abnormal videos. Clustering approaches have focused on modeling sparsity (Lu et al., 2013), enforced high distribution variance between normal and abnormal video representations (Li et al., 2021b), combed dense/sparse clusters for normal/abnormal segments (Zaheer et al., 2020a), and used pseudo labels for anomalous segments (Zaheer et al., 2020b). Another set of methods

(Zhong et al., 2019; Purwanto et al., 2021) has included graph networks to sequentially detect abnormal segments. More recent methods have distinguished between normal and anomalous states with the use of additional modalities such as audio (Wu et al., 2020a) and language context (Yang et al., 2024c; Zanella et al., 2024).

Open-set. As close-set solutions can only model abnormalities in labeled data, models cannot effectively generalize to distributions different than those seen during training. This issue has been studied by Zhao et al. (2011) and Luo et al. (2017) as a sparse-coding (Lee et al., 2006) problem in which the model is trained to reconstruct only plausible normal sequences. Abnormalities are then inferred by large reconstruction loss offsets. Temporal regularity can also be modeled with autoencoders as a reconstruction task (Hasan et al., 2016). To deal with the scarcity of abnormal sequences during training, a number of autoencoder (AE)-based approaches use pseudo representations to improve the embedding space (Astrid et al., 2021b, a). Park et al. (2020) learned prototypes of normal sequences which can then be used to contrast query videos. Other works have constrained the representation space of normal sequences by optimizing piecewise linear decision boundaries (Wang & Cherian, 2019). Two-stream AE frameworks (Cho et al., 2022; Nguyen & Meunier, 2019) have been used to separately reconstruct the appearance and motion characteristics of normal sequences. Generative approaches (Micorek et al., 2024) have recently focused on the latent space with Gaussian Mixture Models (GMM) and inferred an anomaly score across all noise levels. Fioresi et al. (2023) has explored cross-frame mutual information minimization in tandem with a generative objective for privacy preservation. Other privacy-aimed approaches have studied trajectory-based video anomaly detection. Morais et al. (2019) used an RNN to track skeletal points and regressed future locations. The objective of the model was to learn a fixed interpolation for normal sequences with abnormalities captured by the large reconstruction error. Subsequent approaches have used a similar objective to train graph networks (Markovitz et al., 2020), probabilistic models (Flaborea et al., 2023), and masked autoencoders (Stergiou et al., 2024).

5.4.3 Future outlooks

Although great progress has been made, most VAD methods still require adjusting the definitions of normal sequences to update predictions. A direction for future works can be a unified framework that can efficiently adapt predictions through **life-long learning**. Recent works (Yang et al., 2024c; Zanella et al., 2024) fused frame-wise language descriptions from LLMs without requiring re-training the visual model. This can be an effective strategy for upcoming VAD works. Most VAD approaches are also based on the accuracy of

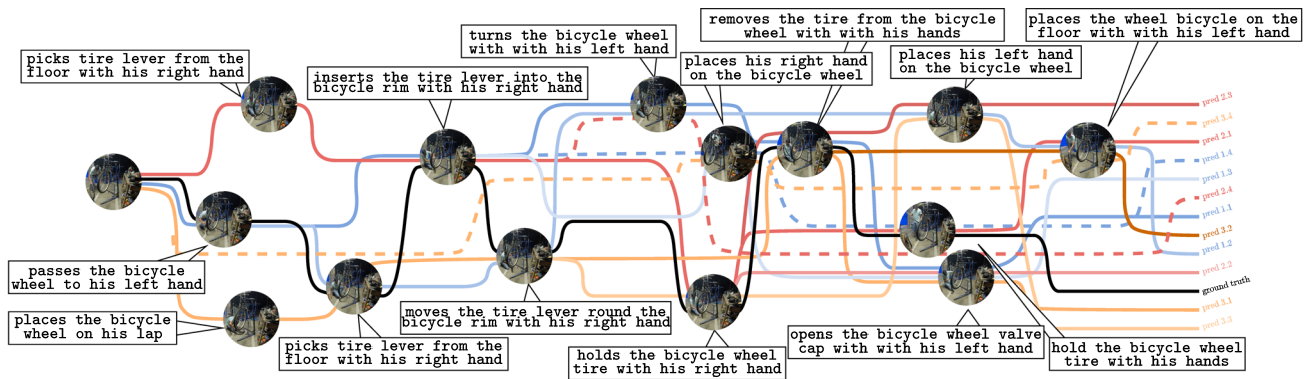


Fig. 14 Forecasting future actions. Starting from the observed action, anticipation approaches infer the sequence of probable next actions. Predictions are shown in a narrative chart format similar to Randall (2009). Example selected from Grauman et al. (2024)

the produced summaries with video-language misalignment directly impacting performance. Potential improvements may explore **in-context learning**. Zhao et al. (2024a) showed that progressively increasing task difficulty through prompts can aid the generalization ability of models to unseen visual scenes.

6 Future forecasting

The future is often uncertain. As shown in Figure 14, a sequence of steps can lead to multiple possible scenarios. Anticipation models are trained on objectives that require the discovery of domain-specific knowledge to address future anticipation challenges. We discuss methods for anticipating categorical semantics in Section 6.1. We then explore approaches for generating future actions in videos in Section 6.2.

6.1 Action anticipation

Action Anticipation (AA) uses current action(s) performed at τ_1 to forecast *proceeding actions* at τ_2 . In contrast to the partial observations for EAP, anticipation tasks only rely on the expected sequence with which actions can be performed. Early works (Kitani et al., 2012; Kuehne et al., 2014; Koppula & Saxena, 2015) have used graphs to model the sequential nature of actions over time. However, to address the long-range dependency limitations of graph-based approaches, works have focused on the procedural execution of actions (Abu Farha et al., 2018; Furnari & Farinella, 2019; Ke et al., 2019), the motion transition intensity between actions (Huang & Kitani, 2014), as well as gaze and hand information (Shen et al., 2018), while defining future-action objectives with multiple predictions (Furnari et al., 2018; Zatsarynna et al., 2024). Despite the diversity in approaches, some challenges remain.

6.1.1 Challenges

Anticipation models predict future actions in sequences. Thus, **future action predictions are accumulated** across multiple rounds. As the future may be unpredictable, errors in these predictions will also influence and reduce the quality of subsequent predictions as the sequence's length increases. Although some models forecast the entire sequence (Gong et al., 2022a; Nawhal et al., 2022), their temporal context is limited compared to autoregressive approaches.

Current works rely on **fixed anticipation time intervals** in which the intermission time $\tau_{1 \rightarrow 2}$ remains constant in training and inference. This significantly limits the applicability of methods in real-world scenarios in which the duration of intervals varies, requiring models to adjust predictions based on conditions such as the speed of execution, difficulty of the action, or the actor's expertise. The majority of existing methods are bound to re-training to accommodate such characteristics.

6.1.2 Anticipation approaches

We discuss three action- and object-based future anticipation approaches.

Embedding similarity maximization. Representations of future actions can be used as targets for learned embeddings. A large number of methods have thus used future embedding reconstruction tasks to infer future action labels. Gao et al. (2017b) used a recurrent decoder to regress future embeddings with an additional policy for class predictions over time. Interactions between objects and actors (Sun et al., 2019c; Luc et al., 2018) have been explored in early works. Subsequent methods aimed to either maximize the similarity between future and current embeddings through memory banks (Liu & Lam, 2022), optimize latent representations for intended goals (Roy & Fernando, 2022), learn prototypes (Diko et al., 2024), or use adversarial representa-

tions (Gammulle et al., 2019). Other generative approaches use pose information as priors (Villegas et al., 2017) or focus on the extrapolation of activity trajectories (Chi et al., 2023). Autoregressive approaches have recently shown great promise using either contrastive objectives (Wu et al., 2020c), causal attention (Girdhar & Grauman, 2021), or audio-visual inputs (Zhong et al., 2023c). As future predictions depend on the usefulness of current observations, works have also integrated uncertainty terms in their predictions. Vondrick et al. (2016a) regressed towards multiple plausible future embeddings, Abdelsalam et al. (2023) grounded the sequentiality of visual embeddings to language, while Guo et al. (2024a) defined probabilistic transformer outputs through a top-k prediction loss similar to Furnari et al. (2018).

Long-term anticipation. The anticipation of the future can also be extended to forecasting multiple upcoming actions over a longer temporal duration. Bokhari and Kitani (2017) used a q-learning framework with reward functions for the activity label, and locations where actions are performed. Nawhal et al. (2022) used a two-stage approach to first infer potential labels and then utilize their logits alongside visual features to predict future action segments. Similarly, Gong et al. (2022a) used learnable latent representations for the future embeddings and cross-attended (Jaegle et al., 2021; Lee et al., 2019) them with the observed video embeddings. Generative approaches have also learned future embeddings from pre-defined temporal states (Piergiovanni et al., 2020), logit sequences (Zhao & Wildes, 2020), cyclic consistency (Abu et al., 2021), or the expected variance in future representations (Mascaró et al., 2023; Patsch et al., 2024). Recently, Mittal et al. (2024) used general language and visual queries to infer prediction through LLMs.

Next active object. A recently introduced set of anticipation tasks concerns the study of object-centric future forecasting. Next active object anticipation forecast the objects that will be used in future actions and has been addressed using predictions on the salient regions (Dessalene et al., 2021), hand position generated representations (Jiang et al., 2021), or autoregressively attending object and visual information (Thakur et al., 2024). Other methods may also forecast human-object interaction regions (Liu et al., 2020, 2022c; Roy et al., 2024), object relations (Roy & Fernando, 2021; Zatsarynna et al., 2021), or time-to-contact estimates (Mur-Labadia et al., 2024).

6.1.3 Future outlooks

Despite the great advancements and number of methods for each anticipation task, less-studied aspects that enhance the applicability of current models exist. Primarily, although the intention of these approaches is their **deployment in real-time scenarios**, they are still used and evaluated in offline settings. Aspects such as latency and inference times for

these models are largely overlooked with only a limited number of works designing stream-based models (Furnari & Farinella, 2022; Girase et al., 2023). The application of these models in real-world settings also requires **prediction adjustability** as inferring labels or sentences of a specific semantic hierarchy may not always be possible. Instead, the prediction granularity needs to be both adjusted and subsequently refined given the available video information. A greater research challenge concerns **multi-person anticipation** in which multiple atomic and group actions need to be forecasted. This also includes forecasting actions in social scenarios in which human-human interactions also need to be predicted, possibly with regard to the social context such as interpersonal relations and roles.

6.2 Video Generation

Forecasting future actions can be extended from the semantic space to the pixel space. Generating real-world physical phenomena includes a high level of complexity (Finn et al., 2016). Capturing simple state transitions of pixels at times does not suffice in learning complex spatiotemporal variations of motions (Wu et al., 2021b). Several types of models have been used throughout the years to address complex scene dynamics.

6.2.1 Challenges

Generating representative future actions is challenging. We visualize three prevalent challenges in video generation in Figure 15.

Consistency failures. Similar to forecasting the semantics of later actions, generative approaches can accumulate errors resulting in a degradation of the frame quality as the video progresses. Methods that either use memory banks (Oshima et al., 2024), coarse to fine representations (Yin et al., 2023a), or generate long sequences in parallel (Zhuang et al., 2024) either require long inference times or are computationally expensive.

Physics failures. As scene dynamics and characteristics of the real world are learned implicitly by generative models, out-of-distribution actions or motions may not be effectively synthesized. Most of the metrics and objectives used to quantify model performance are based only on the visual quality of the generated video. However, the scene's realism also extends to feasible motions, actions, and permutations.

Video-prompt alignment failures. Conditional generation is a recent research topic of high interest. Text-to-Video (T2V) synthesis relies on binding LLM and noise latent embeddings to control video generation. Misalignments in the joint embedding space will be reflected in the outputs.

size past or future frames (temporal outpainting) and current frames (inpainting) of videos based on both cues and textual descriptions. The model's pipeline is shown in Figure 16. A partial video is used alongside a codebook of latents to generate the remaining frames. Stop gradients are used to contrastively update the encoder and codebook. A final frame-wise feature-matching function is used to improve the embedding distance of real and generated frames.

Despite the progress in video generation by adversarial models, instability in training and mode collapse are the primary disadvantages of GANs when generating realistic and diverse videos.

Probabilistic models for video generation. Denoising Diffusion Probabilistic Models (DDPMs) (Ho et al., 2020; Sohl-Dickstein et al., 2015; Song & Ermon, 2019) combine two Markov processes with the first (forward) corrupting the input data to noise within a distribution. The second (backward) process reverses this effect by reconstructing an input from the noisy representation. New inputs not in the training data are generated by sampling the prior distribution. Ho et al. (2022b) directly extended this formulation to video by extending the original U-Net's (Salimans et al., 2017) dimensionality used in the backward step with space-time kernels. Following works (He et al., 2022c; Hong et al., 2022b; Blattmann et al., 2023) have moved away from pixel-level diffusion. They instead utilize the semantically rich and lower-dimensional latent space (Rombach et al., 2022) with autoencoders to encode and project video inputs and outputs. The computational efficiency of Latent Diffusion Models (LDM) has enabled a new stream of works to improve temporal alignment of frames (Blattmann et al., 2023; Yang et al., 2023d) and minimize training data requirements (Nikankin et al., 2023; Wu et al., 2023b). Yu et al. (2023c) combined the two approaches by projecting videos to triplane representations. Yu et al. (2024b) adapted image-based models by incorporating low-resolution temporal content latents computed as the weighted sum of frames. Both image-based and low-resolution motion-based models are denoised with a similar training objective conditioned on the context vector. Although video context can improve frame generation, the number of frames these models can generate is still limited.

Text-conditioned generation. Language embeddings are increasingly used as a prior for video generation. The objective of these methods is to generate videos from textual descriptions based on visual-language correspondence. Dorkenwald et al. (2021) used the start and end times as generation-controlling factors. Following methods explored T2V generation based on VQVAE codebooks conditioned on language (Han et al., 2022; Yan et al., 2021), or language and motion (Hu et al., 2022d). As LDM approaches rely on latent representations, unified visual-language embedding spaces can also be used to generate videos. LDM methods include joint conditional generation of images and videos (Gupta et

al., 2023), shifting latent features for parameter-free temporal variance (An et al., 2023), and concatenating frames in spatial grids (Lee et al., 2024). Zeng et al. (2024) showed that first generating the start and end action states enables models to effectively generate the transitioning frames limiting the dependence on well-formed textual descriptions. A number of recent works (Fei et al., 2024a; Tian et al., 2024b; Wang et al., 2023i, 2024d; Wei et al., 2024; Zhuang et al., 2024) have employed adapters on image-based LDMs similar to ControlNet (Zhang et al., 2023c). Given a pre-trained model using input latents and frozen parameters, a copy of the block is created with trainable parameters to fuse conditional latents of any modality type. These are integrated into the frozen model with projection or cross-attention layers.

Generating long sequences. Generating long videos is challenging as it requires models to learn long-range temporal dynamics. Initial efforts aimed to mitigate reductions in the generation quality over time through hybrid training schemes (Brooks et al., 2022) and by concatenating frame-wise codecs over time for frame consistency in generation (Skorokhodov et al., 2022). Shen et al. (2023a) extended these approaches by using learnable latent vectors to represent motion styles as priors to generate frames. Harvey et al. (2022) explored the conditionality between sampled frames for generating 25min videos with fixed backgrounds. Approaches (Ho et al., 2022a; Singer et al., 2023) have also focused on super-resolution models in tandem with LDMs to generate low-resolution long video sequences which are upsampled to higher resolutions in subsequent steps. Several models have been based on autoregression to accommodate future unpredictability and large video changes. Weissenborn et al. (2020) extended the patch-based generation approach of Subscale Pixel Networks (SPNs) (Menick & Kalchbrenner, 2019) to spatio-temporal voxels. Ge et al. (2022a) used an autoregressive transformer to generate latent representations for the next frames with a VQVAE as a backbone generator. Other auto-regressive VQVAE-based works explored dimension-specific (Wu et al., 2021a) and local attention (Liang et al., 2022b; Wu et al., 2022a). More recently, video generation methods have adapted causal attention encodings fused with previous frame features (Yan et al., 2023c; Villegas et al., 2022), used cross-attention adapters to include temporal context in image generators (Long et al., 2024), and guided the generation with foreground masks (Chang et al., 2024).

6.2.3 Future outlooks

Despite the recent substantial advancements of generative models, rudimentary challenges still exist that, in turn, provide opportunities for future approaches. Crucially, despite the high appearance quality of current models, a standardized **evaluation and benchmark method** is missing. Approaches

based on T2V generation have been shown to replace or fuse concepts, and generate irrelevant objects using specific low-confidence prompts (Du et al., 2023). The scope of current evaluation and benchmark methods (Huang et al., 2024e; Liu et al., 2024f, 2023c; Saito et al., 2020; Unterthiner et al., 2019) is limited to primarily comparing the divergence of generated and real data distributions. Such comparisons do not reflect the extent to which the generated video conforms to the query, and how plausible the output is. The design of domain and characteristic-specific objectives is an interesting research direction.

Another point of improvement for future works is the implementation of **generation control based on physical realism**. Although control of the generation process has been explored in many modalities (Zhang et al., 2023c), the number of works that aim to impose modality-specific characteristics and dynamics constraints in the generation remains small. Video consistency and physics failure cases highlighted in Figure 15 can be addressed by conditional terms that provide implicit information about the visual world. Such information could also be used explicitly by relying on physics simulations that reason about physical objects in the scene (Liu et al., 2024d).

The large capacity of generative models has shown great capabilities in simulating complex scenes. The impact of this has been shown in works that can simulate interactions of actors and objects both in the physical world (Yang et al., 2023c) and virtual renderings (Alonso et al., 2024; Valevski et al., 2024). Understanding aspects of the world to generate video frames aligns with a number of downstream tasks that can enable **generative models to be used as general-purpose models**.

7 Research directions to explore

Progress in video understanding is fast-paced. As shown in Figure 17, vision and language, video generation, and anomaly detection tasks have experienced significant interest in recent years. In tandem, well-established problems such as temporal tasks have remained relevant. We provide a look into the future and explore three main directions of progress beyond the continuation of current trends. We envision ways for future models to reason about abstractions in Section 7.1. We then consider the tasks and objectives that future action understanding models will address in Section 7.2. Finally, we discuss efficiency improvements for training and deployment in Section 7.3.

7.1 Reasoning semantics

With the shift from visual to semantic pattern extraction, the notion of abstraction levels will become more central. We

explore future directions for interpreting actions, considering intentions and goals, and adapting to unseen scenarios.

7.1.1 From action to understanding

Increasingly, action understanding is concerned not only with what is visually depicted but with reasoning about how the depiction is just one out of a multitude of possible perspectives. While action recognition tasks have driven a significant amount of progress on visual representations, future tasks will require more semantic interpretation. When moving from isolated clips of actions to longer episodes depicting behaviors, modeling long-range temporal dependencies becomes more important. Understanding behavior over time requires more than just aggregated interpretations of brief clips. Simultaneously, the distinction between visual observation and interpretation will become weaker, achieving a less deterministic view of action understanding with potentially multiple possible interpretations. Consequently, the automated analysis of videos will shift from objectively measuring or labeling, to a more subjective, context-dependent interpretation. In turn, this will require novel ways of training and evaluation, for example by including humans (Kaufmann et al., 2023).

One perspective on context is to include the intentions of those depicted. Despite VLMs' great progress in learning procedural steps in tasks through natural language-guided embeddings (Li et al., 2024h,i; Wu et al., 2024a), their reliance on visual information remains partial. Al-Tahan et al. (2024) showed that scaling models and data sizes do not offer substantial reasoning performance gains for vision tasks despite strong performance in skill-based tasks. Thus, a new avenue for future approaches is the design of open-world models from multi-level semantics that reflect human intentions and goals. Capturing information about the visual world in a scene may not necessarily require to be associated with language embeddings. Instead, relations could correspond to different pairs or groups of modalities adaptively. Longitudinal data specific to individuals may also be used to tailor the model's understanding of the world based on the user's goals, objectives, intentions, and interactions. Such treatment gives rise to a person- or case-specific perspective, bringing general action understanding to the consumer.

7.1.2 Novel problem adaptation

Zero-shot performance has improved significantly over the past years. This is especially evident in language- and semantic-based video tasks aided by LLMs' large capacity and context. However, limitations remain in tasks orthogonal to pretext SSL (Liu et al., 2024g). Recent approaches such as modality and probabilistic adapters (Chen et al., 2024c; Lin et al., 2023d; Sung et al., 2022; Upadhyay et al., 2023;

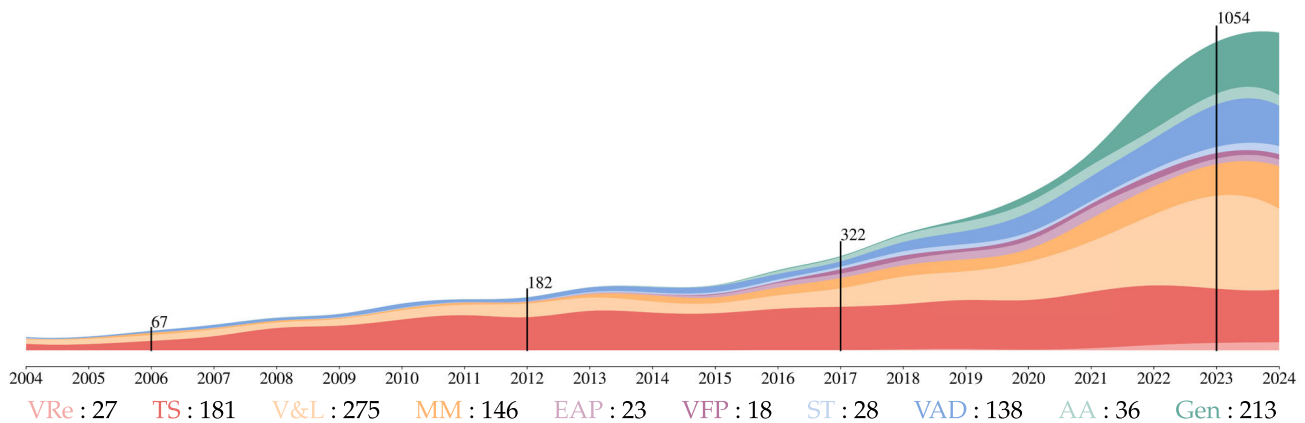


Fig. 17 Number of action understanding papers per year. The research focus (bottom to top) includes video reduction approaches **VRe**, temporal tasks **TS**, vision and language methods **V&L**, multimodal models **MM**, early action prediction **EAP**, video frame prediction **VFP**, state-based tasks **ST**, video anomaly detection **VAD**, action anticipation **AA**,

and video generation **Gen**. The number of relevant papers is approximated from all works citing influential papers with ≥ 300 citations per group (* From Google Scholar as of the 28th of October 2024). Small disparities are expected as recent works may not be included. The increasing research activity in action understanding is evident

Lu et al., 2024a), information gating (Zhang et al., 2024b), visual prompt learning (Khattak et al., 2023), knowledge distillation (Mistretta et al., 2024), and model caching (Zhang et al., 2021d) have improved zero-shot downstream task performance by adjusting pre-trained models. However, only a few structural elements or objectives in models explicitly improve zero-shot performance for unseen distinct tasks. Unified models that can be used as mixture of experts controllers (Bao et al., 2022; Lin et al., 2024; Wang et al., 2022c; Yu et al., 2024a) are promising to bridge this gap. Sparsely trained models in mixture of experts settings allow for faster inference times with only task-relative sub-models using conditional computations (Bengio et al., 2013b; Jacobs et al., 1991). When such mixtures can be linked to different levels of abstraction, re-use of models across levels also becomes feasible. The integration of experts can be done regardless of the backbone architecture.

7.2 Better task definitions

Training models does not necessarily guarantee that temporal dynamics of videos are learned at a foundational level. Future research may revisit and integrate beneficial properties for representation learning. Additionally, future works can explore performance measurements beyond simple metrics, instead focusing on explanation by observing embedding distributions and feature correspondences.

7.2.1 Objectives

Distributions learned from representation-based objectives can be effectively used as priors to downstream tasks (Janocha & Czarnecki, 2017; Larochelle et al., 2009). Draw-

ing inspiration from (Bengio et al., 2013a), a number of widely-accepted target properties are discussed below.

Temporal and spatial coherence. Temporally or spatially proximal instances should correspond to similar representations. This notion can be extended to maintain proportional distances across both the pixel and embedding spaces. This coherence prior has been explored in objectives relating to time such as cyclic consistency (Dwibedi et al., 2018; Donahue & Elhamifar, 2024; Haresh et al., 2021), video procedural learning (Chen et al., 2022c; Sermanet et al., 2018), DTW (Dvornik et al., 2021; Hadji et al., 2021), and cross-frame stochasticity (Zhang et al., 2023b). These priors can be explored in a more general context as pre-training tasks similar to SSL while being tailored to the nature of videos.

Abstractions and hierarchies. Beyond fine-grained categories and semantics, most current works do not explicitly learn levels of abstraction. They are primarily limited to implicit connections between specific types (Li et al., 2024f) that often lead to spurious correlations (Chen et al., 2020b; Kim et al., 2023; Tian et al., 2024a) as well as task- and instance-based misalignment (Zhang et al., 2024e). Objectives that enforce abstraction hierarchies can potentially mitigate such misalignments. Promising efforts include partial order relations (Alper & Averbuch-Elor, 2024), prototype learning (Ramesh et al., 2022), hyperbolic representations (Mettes et al., 2024), and scene graphs (Li et al., 2024b). As models become more polysemantic, the use of natural hierarchies and abstractions is expected to become more prevalent.

Natural clustering and manifolds. Local representations tend to preserve similar polysemantic characteristics. Several works have shown that real data are not represented within the totality of the feature space but instead form dense concentrations in specific regions (Genovese et al., 2012; Jiang

et al., 2018; Liang et al., 2022d). Using the tangent space of these distributions as a prior has shown promise in vision tasks such as generation (He et al., 2023), model explanation (Bordt et al., 2023), anomaly detection (Shin et al., 2023), and corruption robustness (Chen et al., 2022d). However, using the tangent space from real data distributions as an objective-steering prior remains largely an open question for large-scale multimodal action understanding models.

7.2.2 Limitations in performance beyond metrics

Much of the progress in the domain of action understanding originates from comparing model outputs on benchmark data. While reported performance provides insights into the relative merits of models, it does not provide a good understanding of typical failure modes. Recent image-based (Kowal et al., 2024a,b; Park et al., 2023; Walmer et al., 2023) and video-based (Kowal et al., 2024a; Stergiou & Deligiannis, 2023) visualization approaches provide human-interpretable insights into predictions at the instance level. However, understanding how semantic interpretations of actions are addressed, remains largely unexplored. This limits understanding the generalization ability to novel domains and tasks. Uptake of recent explainable AI trends (Minh et al., 2022) into computer vision model development can prompt the development of better measures for the capabilities and limitations of novel models.

Beyond benchmarking models on tasks and metrically evaluating performance, understanding the distributions and learned correlations provides new research opportunities. In-Context Learning (ICL) (Brown et al., 2020; Hoffmann et al., 2022) and Chain-of Thought (CoT) (Wei et al., 2022c) prompting are promising directions for LLMs and VLMs. Bansal et al. (2023) has shown that LLMs' capabilities are influenced by just a small number of attention or feed-forward layers, which are highly task-specific. Both Baldassini et al. (2024); Chen et al. (2024d) showed that ICL in VLMs primarily relies on text information. Disparities between target and learned features can occur due to shortcuts learned by models. Common factors that can lead to shortcuts include contrastive loss' multiple local minima (Robinson et al., 2021), suppression of visual information by language (Li et al., 2023e), and low mutual information between latent representations and real data (Adnan et al., 2022). Recently, Bleeker et al. (2024) showed that introducing unique information distal to the overall training distribution favors VLMs' reliance on shortcuts for models trained on contrastive objectives. Such insights provide opportunities for exploring objectives and models with better multimodal and data-varying generalization capabilities.

7.3 Efficiency

Model efficiency is essential for real-time application. Given the rapid deployment of models in a multitude of applications, we also highlight privacy risks alongside opportunities for domain specialization.

7.3.1 From research to deployment

The increased variety of action understanding tasks also comes with the potential of improving actual deployment. Current and future models achieve performance and robustness levels that allow them to automate processes such as video data curation and surveillance. Novel applications based on behavior analysis can also benefit from these advances. Moving from benchmarks to the real world requires attention to computational efficiency. While the accuracy of current models is remarkable, performance comes at a cost. The trend of increasing model sizes, partly because of the focus on foundation models, largely prohibits the use of these models in computationally constrained operational settings. Attempts to reduce the computational complexity of trained models through pruning (Iofinova et al., 2023), knowledge distillation (Mistretta et al., 2024), or domain-specific adapters (Hu et al., 2021) are not without limitations. The generalization performance gap between the currently best-performing models and those that can run on consumer hardware is significant. Several recent works (Dao et al., 2022; Gu & Dao, 2023; Poli et al., 2023) propose novel processing paradigms that have the potential to scale better. Future work should address whether advances in multimodal training can transfer across both settings and models.

7.3.2 Generalizable priors

Modern models are primarily trained on large-scale uncensored datasets aimed at multi-domain generalization. However, training distributions can include noise or be insufficiently rich for domain-specific datasets. Recent approaches have aimed to reduce training data requirements by including distribution priors at training. Kahana et al. (2022) aimed to improve zero-shot performance with a joint objective that matches label distributions while minimizing the divergence to original zero-shot predictions. Gao et al. (2022a) utilized multiple levels of abstraction to contrast language and visual semantics and improve training efficiency. Nag et al. (2024) used a weakly-supervised approach to refine pseudo-object masks with cross-modal alignment in low-annotation settings. Approaches specifically utilizing priors in videos include concept distillation from normalized language embeddings (Ranasinghe & Ryoo, 2023), and motion-specific alignment between video and textual descriptions of movements (Zhang et al., 2024c). Distilling

learned information to then be used as an optimization prior is a promising route for efficient training by reducing resource requirements. It can also impose a constraint based on the nature of expected motions with potential benefits in model convergence.

7.3.3 Privacy and specialization

Vision-based models are susceptible to attacks that either invert their gradients to reconstruct inputs (Hatamizadeh et al., 2022) or discover intermediate representations (Fang et al., 2023). Such attacks can compromise potentially proprietary training data, and reveal identifiable information. Inputs and features from VLMs can also be inferred through learnable vision-language triggers (Bai et al., 2024a), inference-time adversarial perturbations in frames (Li et al., 2024c), backdoor attacks through adversarial patches in training (Carlini & Terzis, 2022), and injecting malicious prompts during instruction tuning (Liang et al., 2024b). Kariyappa et al. (2023) showed that semantics from the original data can still be recovered even in distributed settings over large batches. Such vulnerabilities can be exploited across downstream tasks. Thus, evaluating model robustness is an important topic that the community should attend to.

The importance of privacy can also be understood through the current shift toward domain-expert sub-models integrated into general-purpose frameworks. Shen et al. (2024) showed that LLM specialization on domain-specific tasks significantly improves zero-shot generalization in related domains. Visual instructional tuning (Bai et al., 2024b) and evolutionary instruction-based prompting (Luo et al., 2024) have also shown promising results for vision-language models. Enhancing video-based models with domain specialization requires further exploration, for example through singular general models of high capacity, or multiple models in holistic frameworks.

8 Conclusion

Video action understanding includes a diverse set of tasks. These previously isolated tasks are increasingly overlapping in terms of the deployed models, utilized training data, and used evaluation protocols. To this end, we provide a comprehensive review of the broad domain of video action understanding. We discussed the main challenges, relevant datasets, and seminal works with an emphasis on recent (multimodal) advancements across tasks, and future research directions. We explicitly included multimodal advances. We focused on three temporal scopes from which tasks and approaches understand actions performed. We discussed recognition tasks that use complete observations of actions to infer fine- or coarse-grained labels. We then overviewed

predictive tasks from partial observations of actions. Finally, we outlined forecasting tasks with anticipation models that infer general scene knowledge and forecast future actions not yet performed. Using time as a stepping stone, we outline current limitations and promising research directions to further advance the scope, robustness, and deployment of action understanding research.

Data Availability We do not use or generate datasets. Dataset statistics used for comparisons are sourced from the respective papers referenced below.

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